

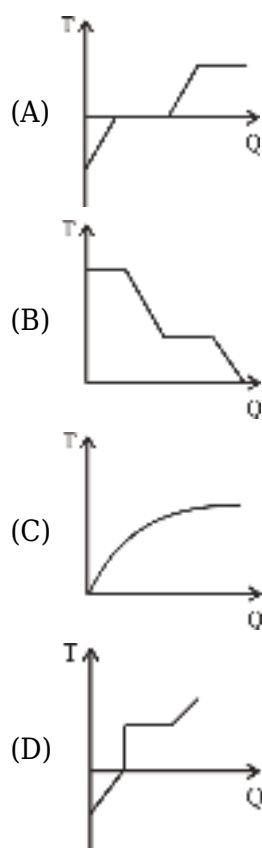
PART 1 : PHYSICS

SECTION-I

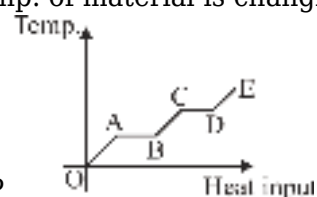
1) 100 g of ice at 0°C is mixed with 100 g of water at 100°C . What will be the final temperature of the mixture

- (A) 10°C
- (B) 20°C
- (C) 30°C
- (D) 40°C

2) A block of ice at -12°C is slowly heated and converted into steam at 100°C . Which of the following curves best represents the event ?



3) A solid material is supplied with heat at a constant rate. The temp. of material is changing with



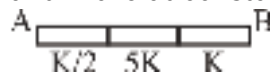
heat input as shown in the figure. What does slope DE represents?

- (A) Latent heat of liquid
- (B) Latent heat of vapour
- (C) Heat capacity of vapour
- (D) Inverse of heat capacity of vapour

4) A sphere and a cube of same material and same volume are heated upto same temperature and allowed to cool in the same surroundings. The ratio of the amounts of radiations emitted will be:-

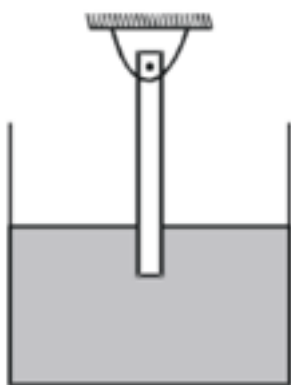
- (A) 1 : 1
- (B) $\frac{4\pi}{3} : 1$
- (C) $\left(\frac{\pi}{6}\right)^{1/3} : 1$
- (D) $\frac{1}{2} \left(\frac{4\pi}{3}\right)^{2/3} : 1$

5) A composite rod made of 3 rods of equal length and cross-section as shown in figure. The end A and B are at constant temperature. The effective thermal conductivity of the bar is :-



- (A) $15K/16$
- (B) $6K/13$
- (C) $5K/16$
- (D) $2K/13$

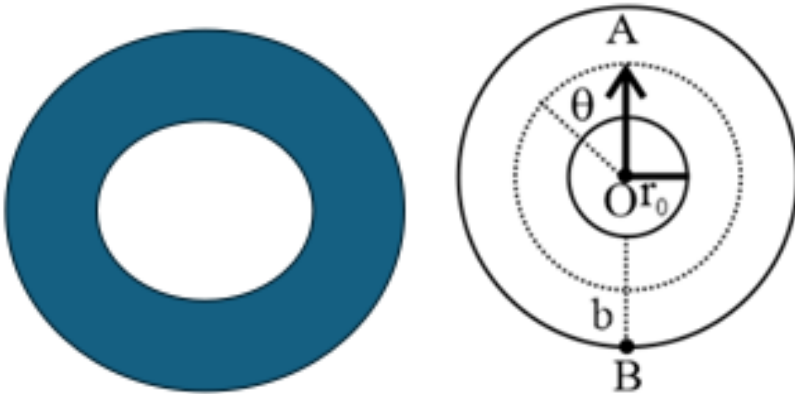
6) A sealed hollow cylinder of plastic is pivoted at the top and partially submerged in liquid as shown. Now if we start rising container slowly then find out the ratio of total length of rod to the submerged length of rod, till its equilibrium will not disturbed. Given density of liquid is $9/5$ times the density of plastic



- (A) 2
- (B) 3
- (C) 4
- (D) 6

7) At Sukhadia Circle, Udaipur (Rajasthan) there is a small circular pond in which let ,water flows at a speed u along a circular path. A Swan shaped boat is steered radially with speed v relative to

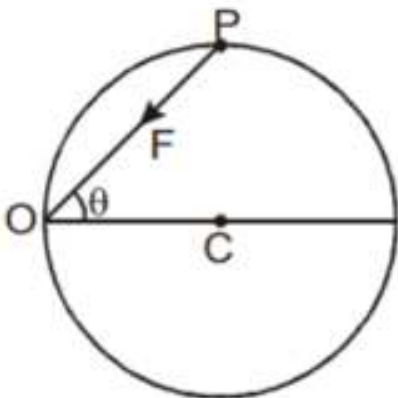
water from inner statue circle (r_0 is radius of central circle.) to reach a diametrically opposite point , as shown in figure, on the other bank in half round (180°). The time to cross the river and reach that point is.



- (A) $\frac{r_0 \left(e^{\frac{v\pi}{u}} - 1 \right)}{v}$
 (B) $\frac{r_0 \left(e^{\frac{v}{u}} - e^\pi \right)}{v}$
 (C) $\frac{r_0 \left(e^\pi - e^{\frac{v\pi}{u}} \right)}{v}$
 (D) None of these

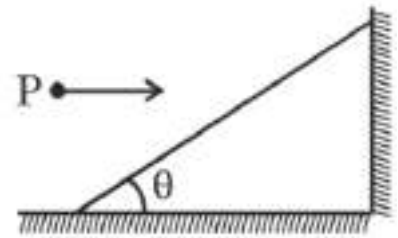
8) As shown in figure a particle of mass ' m ' is moving on the circular path under the influence of a force whose magnitude always remains ' F ' & direction towards a fixed point ' p ' on circumference.

Find the ratio of $\frac{d^2\theta}{dt^2}$ & $\left(\frac{d\theta}{dt} \right)^2$



- (A) $2 \tan \theta$
 (B) $\tan \theta$
 (C) $\frac{\tan \theta}{2}$
 (D) $\frac{\tan \theta}{3}$

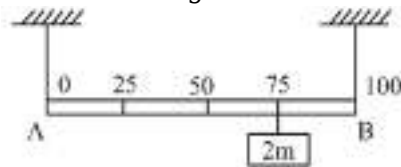
9) In the figure shown a particle $\square$ strikes the inclined smooth plane horizontally and rebounds



vertically. If the angle θ is 60° , then the co-efficient of restitution is:

- (A) $\frac{1}{3}$
- (B) $\frac{1}{\sqrt{3}}$
- (C) $\frac{1}{2}$
- (D) 1

10) Shown in the figure is rigid and uniform one meter long rod AB held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass 'm' and has another weight of mass 2 m hung at a distance of 75 cm from A. the tension in the string at A is :



- (A) 0.5 mg
- (B) 2 mg
- (C) 0.75 mg
- (D) 1 mg

11) The velocity of water in a river is 5 m/s near the surface. If the river is 5 m deep, find the shearing stress between the horizontal layers of water. The co-efficient of viscosity of water = 10^{-2} poise.

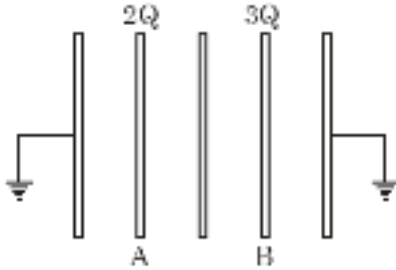
- (A) 10^{-1} N/m^2
- (B) 10^{-2} N/m^2
- (C) 10^{-3} N/m^2
- (D) 10^{-4} N/m^2

12) The primary of a 1 : 3 step-up transformer is connected to a source and the secondary is connected to a resistor R. The power dissipated by R in this situation is P. If R is connected directly to the source will dissipate a power of :

- (A) P/9
- (B) P/3
- (C) P
- (D) 3P

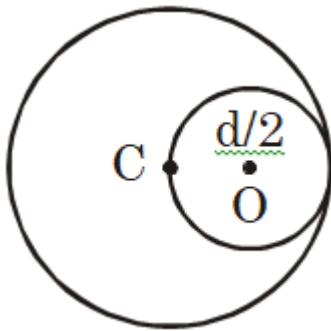
13) In the given figure all identical conductive plates are placed in front of each other all have plate area S and equally separated by d. Plate A and B have been given charges 2Q and 3Q, then find the

potential difference ($V_A - V_B$) in volts between plate B and A. [given that $\frac{Qd}{A\epsilon_0} = 2$ (in SI unit)]



- (A) 1 V
- (B) 2V
- (C) V
- (D) -1V

14) A fixed sphere of radius R and uniform mass density ' d ' has a cavity of radius $\frac{R}{2}$ as shown in figure. The centre of the solid sphere is at ' C ' while that of the cavity is at ' O '. A liquid of density $\frac{d}{2}$ is filled in the cavity. The gravitational force exerted by the liquid on the solid sphere is :-



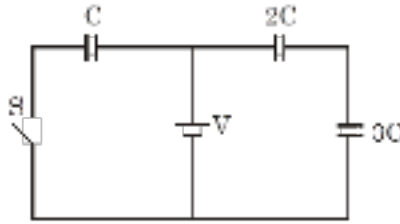
- (A) $\frac{G\pi^2 R^4 d^2}{6}$
- (B) $\frac{G\pi^2 R^4 d^2}{12}$
- (C) $\frac{G\pi^2 R^4 d^2}{9}$
- (D) $\frac{G\pi^2 R^4 d^2}{18}$

15) The angular speed of rotation of earth about its axis when the weight of a man standing on equator will become half of its weight at the north pole (assume earth to be a perfect sphere) $R = 6400$ km.

- (A) $275 \times 10^{-4} \text{ rad sec}^{-1}$

- (B) $9.75 \times 10^{-4} \text{ rad sec}^{-1}$
 (C) $8.75 \times 10^{-4} \text{ rad sec}^{-1}$
 (D) $8.0 \times 10^{-4} \text{ rad sec}^{-1}$

16) In the given circuit diagram, find the heat generated on closing the switch S. (Initially the



capacitor of capacitance C is uncharged) :-

- (A) $\frac{3}{2} CV^2$
 (B) CV^2
 (C) $\frac{1}{2} CV^2$
 (D) $2CV^2$

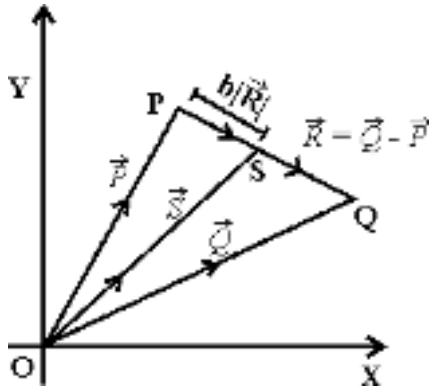
17) Two identical uncharged capacitors A and B are connected in series with an ideal battery. When separation between plates of capacitor A is doubled then choose **INCORRECT** option(s).

- (A) Battery will act as load in this process
 (B) Potential difference between plates of capacitor A increases.
 (C) Potential difference between plates of capacitor B decreases
 (D) Energy of capacitor 'A' increases

18) Two equal positive point charges are kept at points A and B. The electric potential, while moving from A to B along straight line :

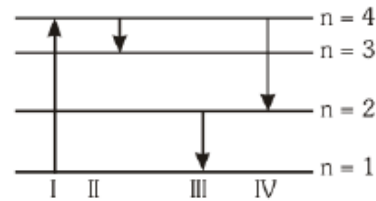
- (A) continuously increases
 (B) remains constant
 (C) decreases then increases
 (D) increases then decreases

19) Three vectors $\vec{P}$, $\vec{Q}$ and $\vec{R}$ are shown in the figure. Let S be any point on the vector $\vec{R}$. The distance between the points P and S is $h|\vec{R}|$. The general relation among vector $\vec{P}$, $\vec{Q}$ and $\vec{S}$ is.



- (A) $\vec{S} = (1-b)\vec{P} + b\vec{Q}$
 (B) $\vec{S} = (b-1)\vec{P} + b\vec{Q}$
 (C) $\vec{S} = (1-b)\vec{P} + b\vec{Q}$
 (D) $\vec{S} = (1-b^2)\vec{P} + b\vec{Q}$

20) The diagram shown the energy levels for an electron in a certain atom. Which transition shown



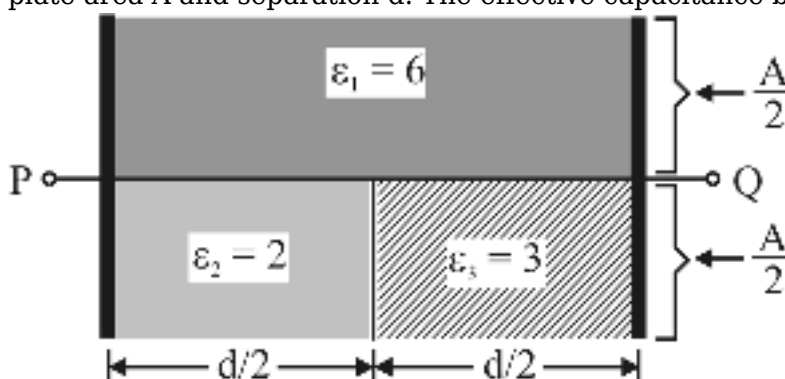
represents the emission of a photon with the most energy ?

- (A) III
 (B) IV
 (C) I
 (D) II

SECTION-II

1) A satellite of the sun is in a circular orbit around the sun, midway between the sun and the earth. Then the acceleration of the satellite is x times the acceleration of the earth. find x (neglect the attraction of earth over satellite).

2) Three dielectric of relative primitivities $\epsilon_{r1} = 6$, $\epsilon_{r2} = 2$ and $\epsilon_{r3} = 3$ are introduced in a parallel plate capacitor of plate area A and separation d . The effective capacitance between P and Q is $\frac{x\epsilon_0 A}{d}$.

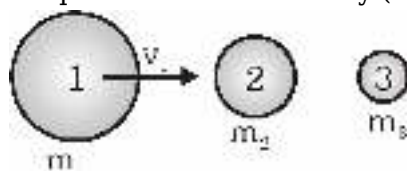


Then $\frac{5}{7}x$ will be.

3) The drag coefficient C_D of an automobile is determined from the expression $C_D = \frac{D}{\frac{1}{2}\rho v^2 S}$ Where D is the drag force experimentally determined in a wind tunnel, ρ is the air density, v is the speed of the air in the wind tunnel, and S is the cross-sectional area of the car presented to the air flow. Determine the dimension of C_D in length.

4) A potentiometer is used for the comparison of e.m.f. of two cells E_1 and E_2 . For cell E_1 the no deflection point is obtained at 20 cm and for E_2 the no deflection point is obtained at 30 cm. The ratio of their e.m.f.'s is $\frac{N}{3}$, then N is:

5) The centres of the spheres 1, 2 and 3 lie on a single straight line. Sphere 1 is moving with an (initial) velocity v_1 directed along this line and hits sphere 2. Sphere 2, acquiring after collision a velocity v_2 , hits sphere 3. Both collisions are absolutely elastic. What must be the mass of sphere 2 (in kg) for the sphere 3 to acquire maximum velocity (The masses m_1 and m_3 of spheres 1 and 3 are 9kg & 1kg respectively)?



PART 2 : CHEMISTRY

SECTION-I

1) Ratio of solubilities of gases N_2 & O_2 in water from air at 25° & 1 atm will be if air is 20 % by volume of O_2 & 80% by volume of N_2

Given : $K_H(N_2) = 2 \times 10^4$ atm

$K_H(O_2) = 10^4$ atm

- (A) 8 : 1
- (B) 1 : 8
- (C) 2 : 1
- (D) 1 : 2

2) The resistance of 0.05 M solution of oxalic acid is 200 ohm and cell constant is 2.0 cm^{-1} , the equivalent conductance (in $\text{S cm}^2 \text{ eq}^{-1}$) of 0.05 M oxalic acid is :-

- (A) 100
- (B) 0.2
- (C) 200
- (D) 400

3) The decomposition of a gaseous substance (A) to yield gaseous products (B) and (C) follows first order kinetics. If initially only (A) is present and 10 minutes after the start of the reaction, the

pressure of (A) is 200 mm Hg and the total pressure is 300mm Hg, then the rate constant for reaction $A \rightarrow B + C$ is

- (A) $(1/600) \ln 1.25 \text{ sec}^{-1}$
- (B) $(2.303/10) \log 1.5 \text{ min}^{-1}$
- (C) $(1/10) \ln 1.25 \text{ sec}^{-1}$
- (D) $\frac{2.303}{600} \log 1.5 \text{ sec}^{-1}$

4) Which of the following reaction is **INCORRECT**?

- (A) Stability order of dihalides $\text{SiX}_2 < \text{GeX}_2 < \text{SnX}_2 < \text{PbX}_2$
- (B) $2\text{NH}_3 + \text{H}_2\text{O} + \text{CO}_2 \rightarrow (\text{NH}_4)_2\text{CO}_3$
- (C) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O} \xrightarrow{T < 373 \text{ K}} \text{Na}_2\text{O} + \text{CO}_2 + 10\text{H}_2\text{O}$
- (D) Boric acid is polymeric due to the presence of hydrogen bonds

5) When H_2S gas is passed in acidic medium which set of cations get precipitated?

- (A) $\text{Mn}^{2+}, \text{Co}^{2+}, \text{Zn}^{2+}$
- (B) $\text{Pb}^{2+}, \text{Cu}^{2+}, \text{Bi}^{3+}$
- (C) $\text{Ba}^{2+}, \text{Sr}^{2+}, \text{Ca}^{2+}$
- (D) $\text{Fe}^{3+}, \text{Cr}^{3+}, \text{Al}^{3+}$

6) On heating compound X a gas Y is obtained, which is also manufactured by reaction of $\text{Ba}(\text{NO}_3)_2 + \text{Zn} + \text{KOH}(\text{aq.})$.

- (A) Compound X is $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$
- (B) Gas Y is used in Ostwald process
- (C) Gas Y is colourless and odourless
- (D) Gas Y produce blue solution in $\text{AgNO}_3 (\text{aq.})$

7) The ΔH° for the mutarotation of glucose in aqueous solution,
 $\alpha - \text{D} - \text{glucose} (\text{aq}) \rightarrow \beta - \text{D} - \text{glucose} (\text{aq})$ has been measured in a microcalorimeter and found to be $-1.16 \text{ kJ} \cdot \text{mol}^{-1}$. The enthalpies of solution of the two forms of glucose have been determined to be
 $\alpha - \text{D} - \text{glucose} (\text{s}) \rightarrow \alpha - \text{D} - \text{glucose} (\text{aq}) \Delta H^\circ = 10.72 \text{ kJ} \cdot \text{mol}^{-1}$
 $\beta - \text{D} - \text{glucose} (\text{s}) \rightarrow \beta - \text{D} - \text{glucose} (\text{aq}) \Delta H^\circ = 4.68 \text{ kJ} \cdot \text{mol}^{-1}$
Calculate ΔH° (in kJ/mol) for the mutarotation of solid $\alpha - \text{D} - \text{glucose}$ to solid $\beta - \text{D} - \text{glucose}$.

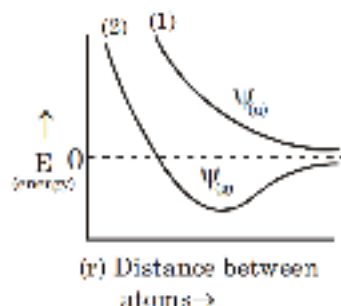
- (A) $+4.88 \text{ kJ/mol}$
- (B) -4.88 kJ/mol
- (C) -2.44 kJ/mol
- (D) $+2.44 \text{ kJ/mol}$

8) Select **INCORRECT** statement regarding KMnO_4 & $\text{K}_2\text{Cr}_2\text{O}_7$.

- (A) Both act as oxidising agent in acidic medium

- (B) Both are paramagnetic and coloured
- (C) O_2 is used in the preparation of both
- (D) The central atom of the anion in both has d^3 hybridisation

9) Select **CORRECT** option regarding formation of molecular orbital from the combination of 1s-1s orbitals.



- (A) Curve-(1) represents formation of BMO
- (B) Curve-(2) represents formation of ABMO
- (C) $\psi_{(g)} = \psi_A + \psi_B$
- (D) $\psi_{(u)} = \psi_A + \psi_B$

10) For a d^6 metal ion in an octahedral field, the correct electronic configuration is :-

- (A) $t_{2g}^6 e_g^0$ when $(\Delta_0 < P)$
- (B) $t_{2g}^4 e_g^2$ when $(\Delta_0 > P)$
- (C) $t_{2g}^6 e_g^0$ when $(\Delta_0 > P)$
- (D) $t_{2g}^3 e_g^3$ when $(\Delta_0 < P)$

11) Which one of the following order is not in accordance with the property stated against it ?

- (A) $Al_2O_3 < SiO_2 < P_2O_5 < SO_3$ (Acidic strength)
- (B) $Al^{3+} < Mg^{2+} < Na^+ < F^-$ (Ionic size)
- (C) $B < C < N < O$ (IE_1)
- (D) $I < Br < F < Cl$ (EA)

12) Both geometrical and optical isomerism shown by :-

- (A) $[Co(en)_2Cl_2]^+$
- (B) $[Co(NH_3)_5Cl]^{2+}$
- (C) $[Co(NH_3)_4Cl_2]^+$
- (D) $[Cr(Ox)_3]^{3-}$

13) In which of the following amino acid 2° -amino group is present :

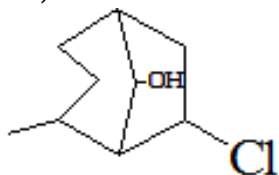
- (A) Glycine
- (B) Glutamine

- (C) Phenyl alanine
- (D) Proline

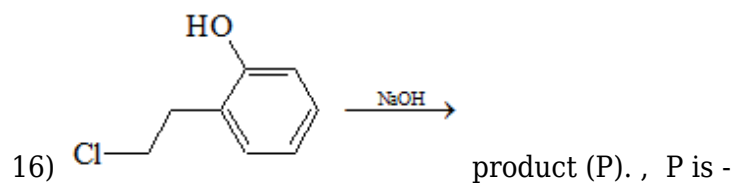
14) Which compound does not react with NaHCO_3

- (A) Squaric acid
- (B) Salicylic acid
- (C) Carboic acid (phenol)
- (D) Acetic acid

15) What will be the correct IUPAC name for the following compound.



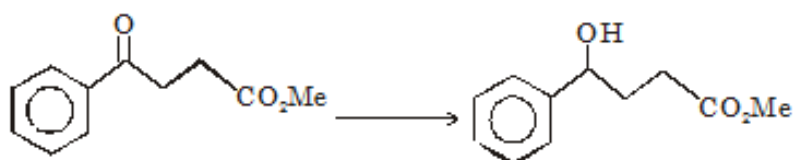
- (A) 7-Chloro-2-methyl bicyclo [3,2,1] octan-1-ol
- (B) 7-Chloro-2-methyl bicyclo [3,2,1] octan-8-ol
- (C) 2-Chloro-7-methyl bicyclo [3,2,1] octan-8-ol
- (D) 2-Chloro-7-methyl bicyclo [3,2,1] octan-1-ol



- (A)
- (B)
- (C)
- (D)

17)

For the following transformation, reagent(s) which can be used is/are :



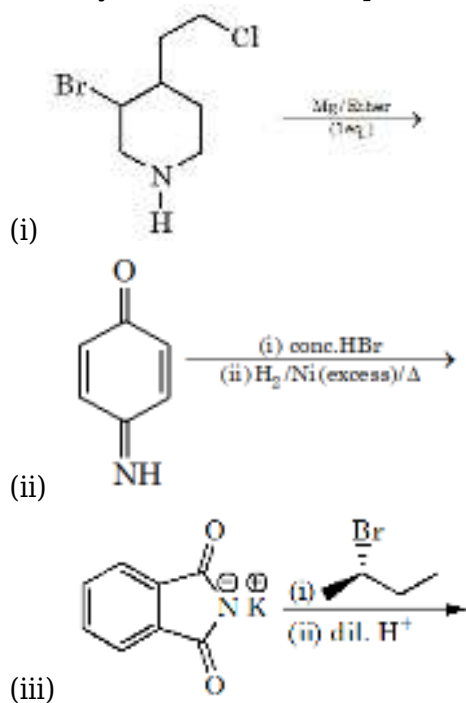
- (A) LiAlH_4
- (B) BH_3 , THF
- (C) H_2 , Raney Ni
- (D) NaBH_4

18) An antipyretic is -

- (A) Seldane
- (B) Paracetamol
- (C) Luminal
- (D) Aspartame

19)

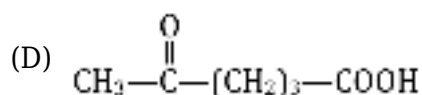
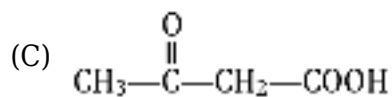
Identify which reaction(s) produce(s) single optically active Amine molecule as product



- (A) i, iii
- (B) iii only
- (C) i, iii
- (D) none of these

20) Which one of the following undergoes decarboxylation most readily?

- (A) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{COOH}$
- (B) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{CH}_2-\text{COOH}$



SECTION-II

1) For the complex $[\text{Pt}(\text{H}_2\text{O})(\text{NH}_3)(\text{Cl})(\text{Br})]$.

If oxidation number of Pt = x

Coordination number = y

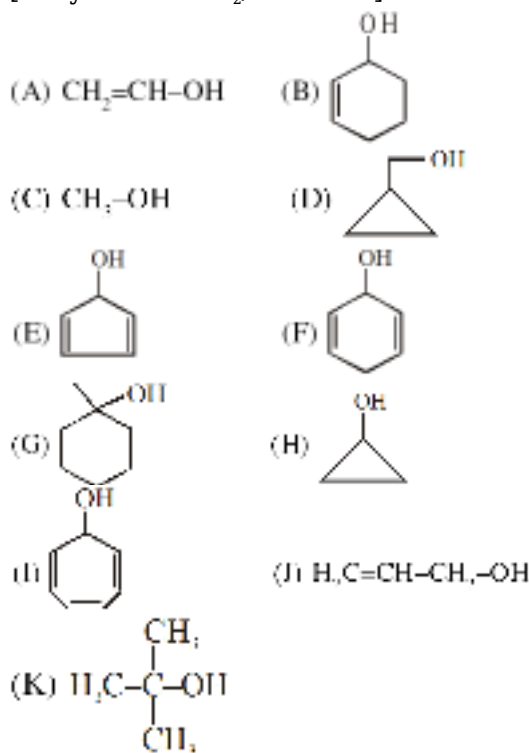
and total possible geometrical isomers = z

then find sum of $x + y + z = ?$

2) Experimentally it was found that a metal oxide has formula $\text{Fe}_{0.98}\text{O}$. Metal Fe, is present as Fe^{2+} and Fe^{3+} in its oxide. Percentage of the metal which exists as Fe^{3+} would be **(nearest integer)** :-

3) 1 mole of monoatomic ideal gas subjected to irreversible adiabatic expansion against constant external pressure of 1 atm starting from initial pressure of 5 atm and initial temperature of 300 K till the final pressure is 2 atm. What is the final temperature (in K) in the process? (take $R = 2 \text{ cal/mol K}$).

4) How many of the following compounds will give immediate turbidity with Lucas reagent [anhydrous $\text{ZnCl}_2/\text{conc. HCl}$].



5) How many of the following compounds, gives positive test for nitrogen in lassaigen's test ?

(i) Hydrazine (ii) Ethanoic acid

- (iii) Ethanamide (iv) Aniline
(v) Hydroxylamine

PART 3 : MATHEMATICS

SECTION-I

1) Solve the equation

$$\cos(3^x \pi) - 2 \cos^2(3^x \pi) + 2 \cos(4(3^x) \pi) - \cos(7(3^x) \pi) \\ = \sin(3^x \pi) + 2 \sin^2(3^x \pi) - 2 \sin(4(3^x) \pi) + 2 \sin(3^{x-1} \pi) - \sin(7(3^x) \pi)$$

- (A) $x = \log_3 \left(\frac{n}{4} + (-1)^n \frac{\pi}{16} - \frac{\pi}{16} \right), n \in \mathbb{Z}$
 (B) $x(2k+1)\pi - \frac{\pi}{5} \cup x = 2m\pi; m \in \mathbb{I}, k \in \mathbb{I}$
 (C) $x = (2k+1)\pi - \frac{\pi}{3} \cup x = 2m\pi, m \in \mathbb{I}, k \in \mathbb{I}$
 (D) $x = (2k+1)\pi - \frac{\pi}{2} \cup x = 2m\pi, m \in \mathbb{I}, k \in \mathbb{I}$

2) Let P_1 be a regular polygon of n sides with side length x and let P_{k+1} be the regular polygon of n sides formed by joining the mid points of consecutive sides of P_k . Also perimeters of the P_i 's form an infinite geometric progression. If $n = 6$ and $x = \frac{1}{2 + \sqrt{3}}$, then the sum of this infinite geometric progression is :

- (A) 8
(B) 17
(C) 14
(D) 12

3) Let $f(x)$ be a non-negative function satisfying $f(x) = \int_0^1 \frac{dt}{f(xt)}$ and $f(1) = 0$ then which of the following is **Not True**

- (A) $f(x)$ is many one function
 (B) $f(x)$ is bounded
 (C) range of the function $f(x)$ is $[0, 1)$
 (D) $f(2) = \frac{1}{2}$

4) Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x (\sec x - \sin x \tan x)}, x \in \left(0, \frac{\pi}{2}\right)$ satisfying the condition $y\left(\frac{\pi}{4}\right) = 2$. Then, $y\left(\frac{\pi}{3}\right)$ is

- (A) $\sqrt{3} (2 + \log_e \sqrt{3})$
 (B) $\frac{\sqrt{3}}{2} (2 + \log_e 3)$
 (C) $\sqrt{3} (1 + 2\log_e 3)$
 (D) $\sqrt{3} (2 + \log_e 3)$

5) If $f(x)$ and $g(x)$ are derivative and antiderivative of a function $h(x)$ where $g\left(\frac{\pi}{2}\right) = 3$ and

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (f(x) + g(x)) \cos x \, dx = 5.$$

Let $p(x)$ be a polynomial of degree 4 such that $p(1) = 7$ and attains its local minimum value 3 at both $x = 2$ and $x = 3$. If the local maximum value of $p(x)$ is equal to λ then the value of $\lambda - g\left(\frac{-\pi}{2}\right)$ is :

- (A) $\frac{19}{8}$
 (B) $\frac{17}{16}$
 (C) $\frac{19}{16}$
 (D) None of these

6) Let $f : (0, \infty) \rightarrow \mathbb{R}$ be continuous function such that $f(x) = 6 (\ln x)^2 - \int_1^e \frac{f(t)}{t} dt$.

Let Δ be area of triangle formed by tangent at $x = \frac{1}{e}$ to curve $y = f(x) + 1$ and co-ordinate axes. Then the value of $\left\lceil \frac{2e\Delta}{3} \right\rceil$ is equal to _____. [where $\lceil . \rceil$ represents G.I.F.]

- (A) 3
 (B) 4
 (C) 6
 (D) 9

7) Let point 'P' be (5, 3) and a point 'R' on $y = x$ and Q on x-axis such that $PQ + QR + RP$ is minimum, then co-ordinates of point Q is -

- (A) (17, 0)
 (B) $\left(\frac{17}{2}, 0\right)$
 (C) $\left(\frac{17}{4}, 0\right)$
 (D) $\left(\frac{23}{4}, 0\right)$

8) The ends A, B of a fixed straight line of length 'a' and ends A' and B' of another fixed straight line of length 'b' slide upon the axis of X & the axis of Y (one end of each line on axis of X & the other end of each line on axis of Y). Find the locus of the centre of the circle passing through A, B, A' and B'.

- (A) $(2ax + 2by)^2 + (2bx - 2ay)^2 = (a^2 - b^2)^2$
 (B) $(2ax - 2by)^2 + (2bx - 2ay)^2 = (a^2 + b^2)^2$
 (C) $(2ax - 2by)^2 + (2bx + 2ay)^2 = (a^2 - b^2)^2$
 (D) $(2ax - 2by)^2 + (2bx - 2ay)^2 = (a^2 - b^2)^2$

9) If $f(x) = \lim_{n \rightarrow \infty} \frac{x^n - x^{-n}}{x^n + x^{-n}}$, $0 < x < 1$, $n \in \mathbb{N}$ then $\int (\sin^{-1} x) f(x) dx$ is equal to

- (A) $- \left[x \sin^{-1} x + \sqrt{1 - x^2} \right] + C$
 (B) $x \sin^{-1} x + \sqrt{1 - x^2} + C$
 (C) $\frac{x^2}{2} + C$
 (D) $\frac{1}{2} (\sin^{-1} x)^2 + C$

10) Let P be (5, 3) and a point R on $y = x$ and Q on the x-axis be such that PQ + QR + RP is minimum. Then the coordinates of Q are

- (A) $\left(\frac{17}{4}, 0 \right)$
 (B) (17, 0)
 (C) $\left(\frac{17}{2}, 0 \right)$
 (D) None of these

11) For $0 < x < \frac{\pi}{2}$, let $f_1(x) = \sum_{r=1}^n \sec \left(x + \frac{r\pi}{6} \right) \sec \left(x + (r-1) \frac{\pi}{6} \right)$ and $f_1(x) - 2f_2(x) = 2 \tan \left(x + \frac{n\pi}{6} \right)$ also $f_2(x) + f_3(x) = 0$ then $f_3(x)$ is

- (A) Discontinuous and non-derivable at $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$
 (B) Neither continuous nor derivative at $x = \frac{2\pi}{5}$
 (C) Continuous and derivable in $\left(0, \frac{\pi}{2} \right)$
 (D) Continuous but not derivable at $x = \frac{2\pi}{5}$

12) Let $f(x) = \lim_{n \rightarrow \infty} \frac{\log(2+x) - x^{2n} \sin x}{1 + x^{2n}}$, then $f(x)$ is discontinuous at:

- (A) $x = 1$ only
 (B) $x = -1$ only
 (C) $x = -1, 1$ only
 (D) no point

13) For $n \in \mathbb{N}$, let x_n be defined as $\left(1 + \frac{1}{n}\right)^{n+x_n} = e$, then $\lim_{n \rightarrow \infty} x_n$ equals

- (A) 1
 (B) $\frac{1}{2}$
 (C) $\frac{1}{e}$
 (D) 0

14) If $\int \frac{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}{\sqrt{\sin^3 x \cos^3 x \sin(x - \theta)}} dx = A\sqrt{\cos \theta \tan x - \sin \theta} + B\sqrt{\cos \theta - \sin \theta \cot x} + C$, where C is the integration constant, then AB is equal to

- (A) $4 \operatorname{cosec}(2\theta)$
 (B) $4 \sec \theta$
 (C) $2 \sec \theta$
 (D) $8 \operatorname{cosec}(2\theta)$

15) Let $f(x)$ be a differentiable function satisfying $e^x \left(\int_0^x f(t) dt + \int_0^x t f(x-t) dt + 1 \right) = 1$ Then the value of $f(2)$ is

- (A) e^2
 (B) $\frac{1}{e^2}$
 (C) $e + e^2$
 (D) $2e^2$

16) Consider two circles C_1 and C_2 as follows:

$$C_1 : x^2 + y^2 + 4x - 13 = 0$$

$$C_2 : x^2 + y^2 - 8x - 13 = 0$$

Let C_1 and C_2 intersect at point A and B. Line L passing through A meets C_1 at P and a line parallel to AP passing through B meets C_2 at Q. Let S be the set of midpoints of line segment PQ as line L varies. Then which is True?

- (A) $(2, 1) \in S$
 (B) $(1, 0) \in S$

(C) $(1, 2) \in S$

(D) $(1, 1) \in S$

17) Let f and g be twice differentiable functions on $\mathbb{R}$ such that

$$f''(x) = g''(x) + 6x$$

$$f'(1) = 4g'(1) - 3 = 9$$

$$f(2) = 3g(2) = 12$$

Then which of the following is **NOT** true ?

(A) $g(-2) - f(-2) = 20$

(B) If $-1 < x < 2$, then $|f(x) - g(x)| < 8$

(C) $|f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$

(D) There exists $x_0 \in \left(1, \frac{3}{2}\right)$ such that $f(x_0) = g(x_0)$

18) If $f(x) = \tan^{-1}x - \frac{2}{\pi}(\tan^{-1}x)^2 + \frac{4}{\pi^2}(\tan^{-1}x)^3 - \dots - \infty$, then the sum of integral values of a for which equation

$$f^2(x) + (\sin^{-1}x)^2 = a$$

, possess solution(s), is

(A) 6

(B) 7

(C) 9

(D) 10

19) If $x = \cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 89^\circ$ and $y = \cos 2^\circ \cos 6^\circ \cos 10^\circ \dots \cos 86^\circ$, then the value of $\frac{2}{7} \log_2(y/x)$, is equal to

(A) 17

(B) 18

(C) 19

(D) 20

20) For $x \geq 3$, Let $f(x) = \frac{-x + x\sqrt{4x-3}}{2}$

Suppose $a_1, a_2, a_3, \dots$ be sequence satisfying $a_1 > 3$ and $a_{2024} = 2024$ and for $n = 1, 2, 3, \dots, 2023$

Define $a_{n+1} = f(a_n)$. Then

$$a_1 + \left(\sum_{i=1}^{2023} \frac{(a_{i+1})^3}{(a_{i+1})^2 + a_i a_{i+1} + (a_i)^2} \right)$$

is

(A) 2024

(B) 4047

(C) 2025

SECTION-II

1) Let $y = f(x)$ ($f : \mathbb{R} \rightarrow \mathbb{R}$) be an explicit function defined by the implicit equation $x^3 + y^3 + 3(x^2 + y^2) + 3(x + y) = 14$ and g be the inverse function of f .

If $\frac{d}{dx} [f(x + g(x)) \cdot g(x + f(x))]$ at $x = -1 + \sqrt[3]{15}$ is equal $\lambda(15)^{\frac{2}{3}}$, where $\lambda \in \mathbb{I}$, then find the value of $|\lambda|$.

2) Let $(a_1, a_2, a_3, \dots, a_n)$ be a permutation of $(1, 2, 3, \dots, n)$ for which $a_1 > a_2 > a_3 > \dots > a_{\frac{n}{2}}$ and $a_{\frac{n}{2}} < a_{\frac{n}{2}+1} < a_{\frac{n}{2}+2} < \dots < a_n$ for n as, an even positive integer. Also $a_1 > a_2 > a_3 > \dots > a_{\frac{n+1}{2}}$ and $a_{\frac{n+1}{2}} < a_{\frac{n+1}{2}+1} < \dots < a_n$ for n as an odd positive integers. Let the total number of permutation of n be $p(n)$, if $200 < p(n) < 500$, then last digit of sum of the values of n is

3) A cricket player played n ($n > 1$) matches during his career and made a total of

$$\frac{(n^2 - 12n + 39)(4 \cdot 6^n - 5 \cdot 3^n + 1)}{5}$$

runs. If T_r represent the runs made by the player in r^{th} match such that $T_1 = 6$ and $T_r = 3T_{r-1} + 6^r$, $2 \leq r \leq n$ then n is :

4)

$$\text{Let } \int (\cot 2x \cot 3x - \tan 2x \tan 7x) \tan 5x \, dx$$

$= a \ln(\tan 2x) + b \ln(\sin 3x) + c \ln(\sec 5x) + d \ln(\cos 7x) + C$ where $a, b, c, d \in \mathbb{Q}$ and C is the

constant of integration. If $(a + b + c + d)$ can be expressed as $\frac{m}{n}$ in the lowest form, then $(m + n)$ equals

5) If

$$\int \frac{\left(\frac{2}{3} + x^2\right) dx}{3 - 4\sqrt{3}x + 9x^4} = \frac{1}{6\sqrt{6}} \tan^{-1} \left(\frac{\sqrt{3}x + 1}{\sqrt{2}} \right) - \frac{1}{6\sqrt{3}} \cdot \frac{1}{f(x)} + C$$

where C is constant of integration and $x \neq \frac{1}{\sqrt{3}}$, $f(0) = -1$, then value of $f(\sqrt{3})$ is

ANSWER KEYS

PART 1 : PHYSICS

SECTION-I

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	A	A	D	C	A	B	A	A	A	D	C	A	D	D	C	C	D	C	C	A

SECTION-II

Q.	21	22	23	24	25
A.	4	3	0	2	3

PART 2 : CHEMISTRY

SECTION-I

Q.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
A.	C	A	A	C	B	B	A	B	C	C	C	A	D	C	B	C	D	B	B	C

SECTION-II

Q.	46	47	48	49	50
A.	9	4	255	7	2

PART 3 : MATHEMATICS

SECTION-I

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	A	D	D	A	B	D	C	D	A	A	C	C	B	D	B	D	B	D	C	B

SECTION-II

Q.	71	72	73	74	75
A.	0	3	6	499	2

SOLUTIONS

PART 1 : PHYSICS

1) **Explain Question:** Find final temperature of mixture of ice and water

Concept: Calorimetry (transfer of energy)

Solution:

$$\theta_{\text{mix}} = \frac{\theta_w - \frac{L_i}{c_w}}{2} = \frac{100 - \frac{80}{1}}{2} = 10^\circ\text{C}$$

Final Answer: (A)

2) **Explain Question:** Graph for ice converting to steam

Concept: Transfer of energy

Solution:

Various process will be as follows :

Ice at -12°C $\xrightarrow{\text{T increase}}$ Ice at 0°C $\xrightarrow{\text{T increase}}$ water at 0°C
 $\downarrow$ T increase
 water at 100°C
 $\downarrow$ T = constant
 steam at 100°C

Final Answer: (A)

3) **Explain Question:** Temperature vs heat graph

Concept: $Q = mC\Delta T$

Solution:

$$(\text{Slope})_{DE} \propto \frac{1}{\text{Specific heat of vapour}}$$

Final Answer: (D)

4) **Explain Question:** Cooling due to radiation

Concept: Thermal radiation

Solution:

$Q = \sigma A t (T^4 - T_0^4)$ If T, T_0 , σ and t are same for both bodies then

$$\frac{Q_{\text{sphere}}}{Q_{\text{cube}}} = \frac{A_{\text{sphere}}}{A_{\text{cube}}} = \frac{4\pi r^2}{6a^2} \quad \dots(i)$$

But according to problem, volume of sphere

$$= \text{Volume of cube} = \frac{4}{3}\pi r^3 = a^3 \Rightarrow a = \left(\frac{4}{3}\pi\right)^{1/3} r$$

Substituting the value of a in equation (i) we get

$$\frac{Q_{\text{sphere}}}{Q_{\text{cube}}} = \frac{4\pi r^2}{6a^2} = \frac{4\pi r^2}{6\left\{\left(\frac{4}{3}\pi\right)^{1/3} r\right\}^2}$$

$$= \frac{4\pi r^2}{6\left(\frac{4}{3}\pi\right)^{2/3} r^2} = \left(\frac{\pi}{6}\right)^{1/3} : 1$$

Final Answer: (C)

5) **Explain Question:** Find effective thermal conductivity

Concept: Equivalent thermal conduction

Solution:

$$R = R_1 + R_2 + R_3$$

$$\frac{3L}{K_{\text{eq}}A} = \frac{L}{\frac{K}{2}A} + \frac{L}{5KA} + \frac{L}{KA}$$

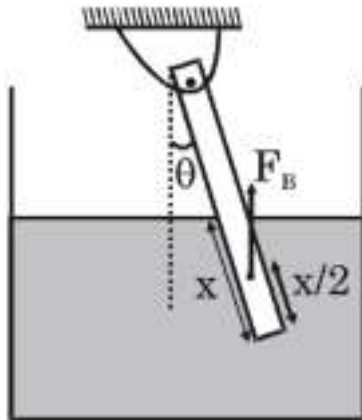
$$K_{\text{eq}} = \frac{15}{16} K$$

Final Answer: (A)

6) **Explain Question:** Find ratio of total length of rod to submerged length till equilibrium

Concept: Bloyancey

Solution:



$$\text{where } F_B = \frac{9\rho}{5} (xA) g$$

Balancing torque about hinge

$$(\rho \ell A) g \times \left(\frac{\ell}{2} \sin \theta\right) = \left[\frac{9\rho}{5} \times (xA)\right] g \times \left[\left(\ell - \frac{x}{2}\right) \sin \theta\right]$$

[where θ is very small]

Final Answer: (B)

7) **Explain Question:** Time to cross the river moving in circle

Concept: Circular motion

Solution:

$$u = \frac{rd\theta}{dt}$$

$$v = \frac{dr}{dt}$$

$$r \frac{d\theta}{dr} = \frac{u}{v} \Rightarrow \int_{r_0}^{r_0+x} \frac{dr}{r} = \frac{v}{u} \int_0^\pi d\theta$$

$$\ln \frac{r_0+x}{r_0} = \frac{v\pi}{u}$$

$$x = r_0 \left(e^{\frac{v\pi}{u}} - 1 \right)$$

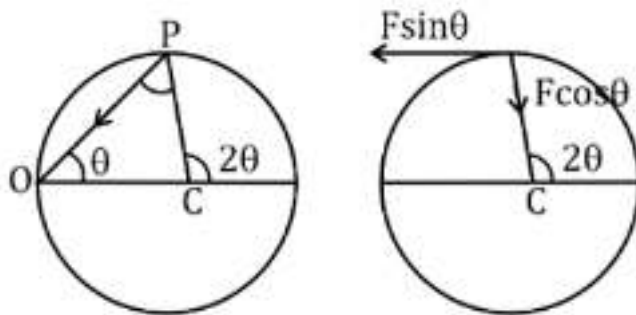
$$t = \frac{x}{v} = \frac{r_0}{v} \left(e^{\frac{v\pi}{u}} - 1 \right)$$

Final Answer: (A)

8) **Explain Question:** Find ratio of $\left(\frac{d\theta}{dt} \right)^2$

Concept: Circular motion

Solution:



$$F \cos \theta = mR\omega^2, F \sin \theta = ma_t$$

Angular velocity ω of line joining P and C is

$$\omega = \frac{d}{dt}(2\theta) = \frac{2d\theta}{dt} \Rightarrow \frac{d\theta}{dt} = \frac{\omega}{2}$$

Tangential acceleration of particle

$$a_t = \frac{F \sin \theta}{m} = R\alpha \Rightarrow \alpha = \frac{F \sin \theta}{mR}$$

$$\therefore \alpha = \frac{d\omega}{dt} = \frac{d}{dt} \left(2 \frac{d\theta}{dt} \right) = 2 \frac{d^2\theta}{dt^2}$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = \frac{\alpha}{2}; \quad \therefore \left(\frac{d^2\theta}{dt^2} \right) = \frac{F \sin \theta}{2mR}$$

$$\Rightarrow \left(\frac{d\theta}{dt} \right)^2 = \frac{1}{4} \omega^2 = \frac{1}{4} \times \frac{F \cos \theta}{mR}$$

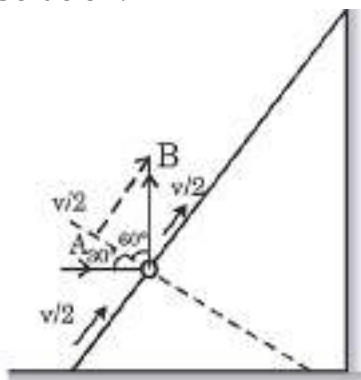
$$\text{SO, } \frac{\left(\frac{d^2\theta}{dt^2} \right)}{\left(\frac{d\theta}{dt} \right)^2} = \frac{\frac{F \sin \theta}{2mR}}{\frac{F \cos \theta}{4mR}} = 2 \tan \theta$$

Final Answer: (A)

9) **Explain Question:** Find co-efficient of restitution

Concept: Collision on wedge

Solution:



Let 'v' be the initial velocity.

Tangential velocity remains same during collision and equal to

$$v \cos 60^\circ = v/2$$

Let v' be the normal component of velocity after impact.

$$\text{In } \triangle OAB : \tan 60^\circ = \frac{v/2}{v'} \Rightarrow v' = \frac{v}{2\sqrt{3}}$$

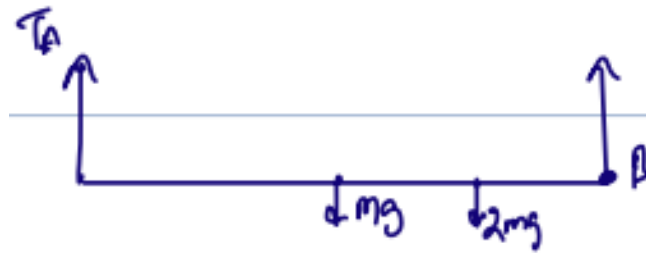
$$\text{Then : } e = \frac{v'}{v \cos 30^\circ} = \frac{(v/2\sqrt{3})}{(\sqrt{3}v/2)} = \frac{1}{3}$$

Final Answer: (A)

10) **Explain Question:** Find tension in the string

Concept: Complete equilibrium

Solution:



$$\begin{aligned} T_A \times 1 &= mg \times \frac{1}{2} + 2mg \times \frac{1}{4} \\ &= \frac{mg}{2} + \frac{mg}{2} = mg \end{aligned}$$

Final Answer: (D)

11) **Explain Question:** Find shear stress of water

Concept: Newton's law of viscosity

Solution:

Velocity of water (du) = 5 m/s

Depth of river (dy) = 5m

Coefficient of viscosity = 10^{-2} poise

$$\therefore 1 \text{ poise} = 0.1 \frac{\text{N.s}}{\text{m}^2}$$

$$\therefore 10^{-2} \text{ poise} = 10^{-2} \times 0.1 = \frac{\text{N.s}}{\text{m}^2}$$

$$\eta = 10^{-3} \frac{\text{N.s}}{\text{m}^2}$$

According to Newton's law of viscosity :-

$$F = \eta A \frac{du}{dy}$$

$$\frac{F}{A} = \eta \frac{du}{dy}$$

$$\tau = \eta \frac{du}{dy} \quad \left(\because \frac{F}{A} = \tau \right) = (\text{shearing stress})$$

$$\tau = 10^{-3} \times \frac{5}{5}$$

$$\tau = 10^{-3} \text{ N/m}^2$$

Final Answer: (C)

12) **Explain Question:** Power dissipation in step up transformer

Concept: Transformer

Solution:

$$\frac{N_2}{N_1} = \frac{V_2}{V_1}$$

$$3 = \frac{V_2}{V_1}$$

$$\frac{1}{V_2} = \frac{V_1}{3V}$$

$$V_2 = 3V$$

Power across 'R'

$$P = \frac{(3V)^2}{R} = \frac{9V^2}{R}$$

$$P_{\text{new}} = \frac{V^2}{R} = \frac{P}{9}$$

Final Answer: (A)

13) **Explain Question:** Find potential difference between plate B and A

Concept: Parallel plate capacitor

Solution:

$$V_A - V_B = -\frac{Qd}{2\Lambda\epsilon_0} = -1V$$

Final Answer: (D)

14) **Explain Question:** The gravitational force exerted by the liquid on the solid sphere

Concept: Gravitational field inside solid sphere

Solution:

$$E = \frac{2Gd\pi R}{3}$$
$$F = ME = \frac{2Gd\pi R}{3} \left[\frac{d}{2} \cdot \frac{4}{3} \pi \left(\frac{R}{2} \right)^3 \right] = \frac{Gd^3\pi^2 R^4}{18}$$

Final Answer: (D)

15) **Explain Question:** Change in weight of person due to rotation of earth

Concept: Variation of g due to rotation

Solution:

$$\frac{g}{2} = g - R\omega^2$$
$$\Rightarrow \omega = \sqrt{\frac{g}{2R}}$$

Final Answer: (C)

16)

Explain Question: Heat generated on closing the switch

Concept: Current flow generates heat

Solution:

Only charge is that capacitor 'C' will get charged.

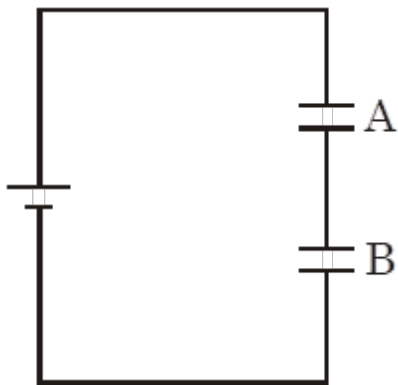
$$\text{Hence heat} = \frac{1}{2} CV^2$$

Final Answer: (C)

17) **Explain Question:** Choose incorrect statement based on capacitors

Concept: Energy of capacitor

Solution:



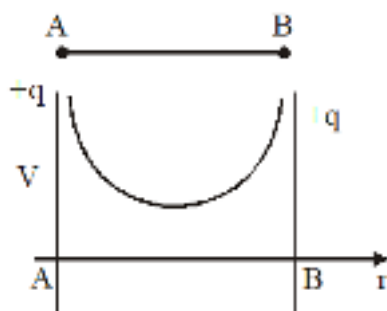
when separation becomes doubles capacitance of A and also equivalent capacitance decreases there for charge on both A and B decreases.

Final Answer: (D)

18) **Explain Question:** Electric potential while moving from A to B

Concept: $V = \frac{kq}{r}$

Solution:



Final Answer: (C)

19) **Explain Question:** Find general relation between $\vec{P}$, $\vec{Q}$ and $\vec{S}$

Concept: Vector addition

Solution:

$$\vec{P} + b\vec{R} = \vec{S}$$

$$\vec{P} + b(\vec{Q} - \vec{P}) = \vec{S}$$

$$\vec{P}(1 - b) + b\vec{Q} = \vec{S}$$

Final Answer: (C)

20) **Explain Question:** Which transition shows emission of photon with most energy

Concept: Energy of photon from transition

Solution:

$$(E_2 - E_1) > (E_4 - E_2) > (E_4 - E_3)$$

When e^- goes from $n = 1$ to $n = 4$

no emission, only absorption

Final Answer: (A)

21) **Explain Question:** Acceleration of satellite as compared to earth

Concept: Gravitational force

Solution:

$$a_E = \frac{GM_S}{r_E^2} \text{ and } a_S = \frac{GM_S}{\left(\frac{r_E}{2}\right)^2} \Rightarrow a_S = 4a_E$$

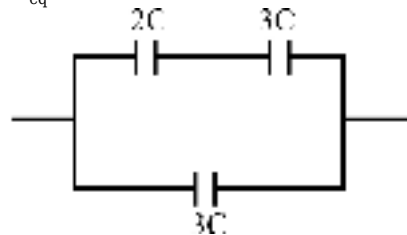
Final Answer: (4)

22) **Explain Question:** Find effective capacitance between P & Q

Concept: Series & parallel combination

Solution:

$$C_{eq} = \frac{21}{5} \times \frac{\epsilon_0 A}{d}$$



Final Answer: (3)

23) **Explain Question:** Find ratio where deflection is 0.

Concept: Potentiometer

Solution:

$$[C_D] = \frac{[D]}{[\rho][V^2][S]} = \frac{[M^1 L^1 T^{-2}]}{[M^1 L^{-3}][L^2 T^{-2}][L^2]}$$

$$= [M^0 L^0 T^0]$$

□ dimension in L is zero

Final Answer: (0)

24) **Explain Question:** Find ratio where deflection is 0.

Concept: Potentiometer

Solution:

$$\frac{E_1}{E_2} = \frac{l_1}{l_2} = \frac{2}{3}$$

Ratio will be equal to the ratio of no deflection lengths i.e.

Final Answer: (2)

25) **Explain Question:** What must be mass of 2 for 3 to after $v_{\max}$

Concept: Collision

Solution:

$$\text{As } v_2 = \frac{2m_1}{m_1 + m_2} v_1$$

$$\square v_3 = \frac{2m_2}{m_2 + m_3} v_2 = \frac{4m_1 m_2 v}{(m_1 + m_2)(m_2 + m_3)}$$

For maximum v_3 , $\frac{dv_3}{dm_2} = 0 \Rightarrow m_2 = \sqrt{m_1 m_3} = 3\text{kg}$

Final Answer: (3)

PART 2 : CHEMISTRY

26) **Explain Question:** Calculate ratio of solubility of N_2 & O_2

Concept: Concept of Henry's law

Solution:

$$X_{N_2} = 0.8, X_{O_2} = 0.2 \text{ in air}$$

$$P_{N_2} = 0.8 \text{ atm}, P_{O_2} = 0.2 \text{ atm}$$

$$P = K_H X \text{ (Henry's law)}$$

$$0.8 = 2 \times 10^4 X_{N_2} \quad \dots\dots(1)$$

$$0.2 = 10^4 X_{O_2} \quad \dots\dots(2)$$

Ratio of solubility of N_2 & O_2 is 2 : 1

Final Answer: (C)

27) **Explain Question:** Calculate equivalent conductance of oxalic acid

$$\text{Concept: } \square \text{ Equivalent conductance} = \frac{K \times 1000}{N}$$

Solution:

$$\text{Specific conductance, } K = \frac{1}{R} \cdot \frac{\ell}{A} = 10^{-2} \text{ S cm}^{-1}$$

$$\square \text{ Equivalent conductance} = \frac{K \times 1000}{N}$$

$$= \frac{10^{-2}}{10^{-1}} \times 10^3$$

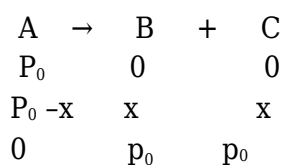
$$= 10^2 \text{ S cm}^2 \text{ eq}^{-1}$$

Final Answer: (A)

28) **Explain Question:** Calculate rate constant for reaction $A \rightarrow B + C$

Concept: Apply concept of first order kinetics.

Solution:



Given :

$$p_0 - x = 200$$

$$p_0 + x = 300$$

$$\Rightarrow p_0 = 250$$

$$x = 50$$

$$k = \frac{1}{t} \ln \frac{p_0}{p_0 - x}$$

$$= \frac{1}{600} \ln \frac{250}{250 - 50} \text{ sec}^{-1}$$

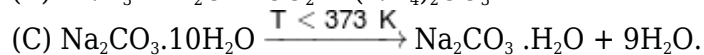
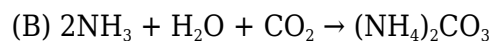
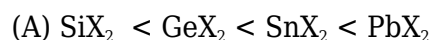
$$= \frac{1}{600} \ln 1.25 \text{ sec}^{-1}$$

Final Answer: (A)

29) **Explain Question:** Identify incorrect statement

Concept: Stability of dihalides of group 14 elements.

Solution:



(D) Boric acid is polymeric due to the presence of hydrogen bonds

Final Answer: (C)

30) **Explain Question:** Identify which set of cations get precipitated by H_2S

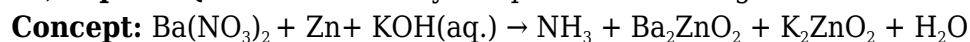
Concept: Sparingly soluble sulphides get precipitated.

Solution:

Sparingly soluble sulphides derived from (Pb^{2+} , Cu^{2+} , Bi^{3+}) ions get precipitated.

Final Answer: (B)

31) **Explain Question:** Identify compound X & Y for given reaction.



Solution:



(Used in ostwald process for the manufacture of nitric acid)

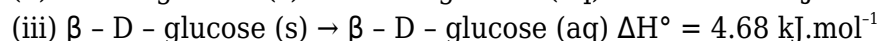
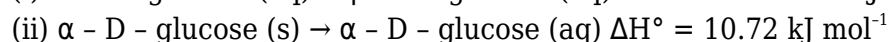
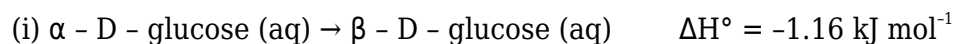
$\text{NH}_3 \rightarrow$ Pungent smell

Final Answer: (B)

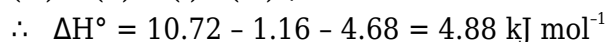
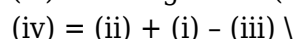
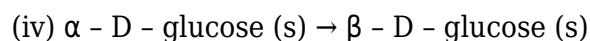
32) **Explain Question:** Calculate enthalpy for the Mutarotation of glucose.

Concept: By using transition reaction obtain the final change, α - D - glucose (s) $\rightarrow$ β - D - glucose (s)

Solution:



To obtain reaction

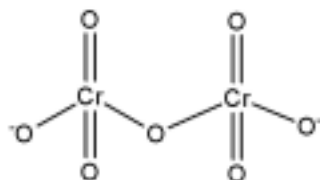
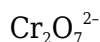
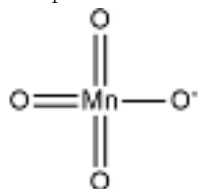


Final Answer: (A)

33) **Explain Question:** Identify incorrect statement

Concept: Chemical & physical properties of KMnO_4 & $\text{K}_2\text{Cr}_2\text{O}_7$

Solution:



- (A) Both act as oxidising agent in acidic medium
- (B) Both are diamagnetic and coloured compound
- (C) O_2 is used in the preparation of both
- (D) The central atom of the anion in both has d^3 hybridisation

Final Answer: (B)

34) **Explain Question:** Identify correct statement regarding formation of molecular orbitals.

Concept: $\psi_{(g)} = \psi_A + \psi_B$ (formation of bonding molecular orbital i.e. linear combination of atomic orbital $\psi_A + \psi_B$)

Solution:

Curve-(2) represents formation of BMO

$\psi_{(g)} = \psi_A + \psi_B$ (formation of bonding molecular orbital i.e. linear combination of atomic orbital $\psi_A + \psi_B$)

Final Answer: (C)

35) **Explain Question:** Find correct electronic configuration of d^6

Concept: No pairing if $P > \Delta_o$

Solution:

$t_{2g}^6 e_g^0$ when $(\Delta_o > P)$

Final Answer: (C)

36) **Explain Question:** Identify the correct order of for given properties

Concept: Concept of acidic strength, ionic radius, IE & EA.

Solution:

(A) Acidity of the oxides of non-metals increases with the electronegativity and oxidation number of the element.

$(Al_2O_3 < SiO_2 < P_2O_3 < SO_3)$

(B) $Al^{3+} < Mg^{2+} < Na^+ < F^-$ (Ionic size)

(C) $B < C < O < N$ (IE_1)

(D) $I < Br < F < Cl$ (EA)

Final Answer: (C)

37) **Explain Question:** Which compound shows both OI & GI

Concept: Apply concept of OI & GI.

Solution:

$[Co(en)_2Cl_2]^+ \rightarrow$ Geometrical and optical isomerism

$[Co(NH_3)_5Cl]^{2+} \rightarrow$ No geometrical isomerism

$[Co(NH_3)_4Cl_2]^+ \rightarrow$ Geometrical isomerism

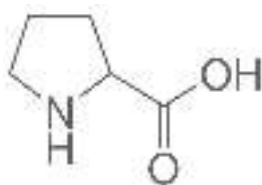
$[Cr(Ox)_3]^{3-} \rightarrow$ Optical isomerism

Final Answer: (A)

38) **Explain Question:** Which amino acid having 2° amine group

Concept: $-NH-$ is 2° amine

Solution:



(Proline)

Final Answer: (D)

39) **Explain Question:** Identify compound which does not react with NaHCO_3

Concept: weak acid than H_2CO_3 hence it will not react with NaHCO_3

Solution:

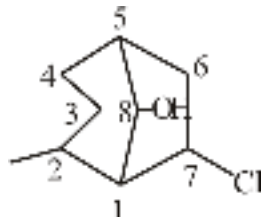
Phenol is weak acid than H_2CO_3 hence it will not react with NaHCO_3

Final Answer: (C)

40) **Explain Question:** Identify correct IUPAC name of given compound.

Concept: Follow IUPAC nomenclature.

Solution:



7-Chloro-2-methyl bicyclo [3,2,1] octan-8-ol

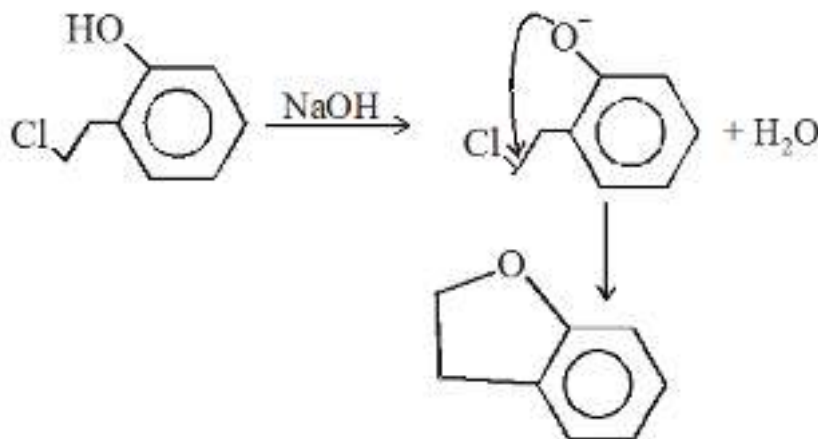
Hence, option (2) is correct.

Final Answer: (B)

41) **Explain Question:** Identify the product 'P'

Concept: Acid - base reaction is followed by intra molecular SN^2

Solution:



Final Answer: (C)

42) **Explain Question:** Which reagent produce the given product from the given reactant.

Concept: NaBH_4 reduces only acyl halide, aldehyde and ketone to alcohol

Solution:

NaBH_4 reduces only acyl halide, aldehyde and ketone to alcohol.

Final Answer: (D)

43) **Explain Question:** Which is an antipyretic drug

Concept: Paracetamol → antipyretic

Seldane → Antihistamine

Aspartame → artificial sweetening agent

Solution:

Paracetamol → antipyretic

Seldane → Antihistamine

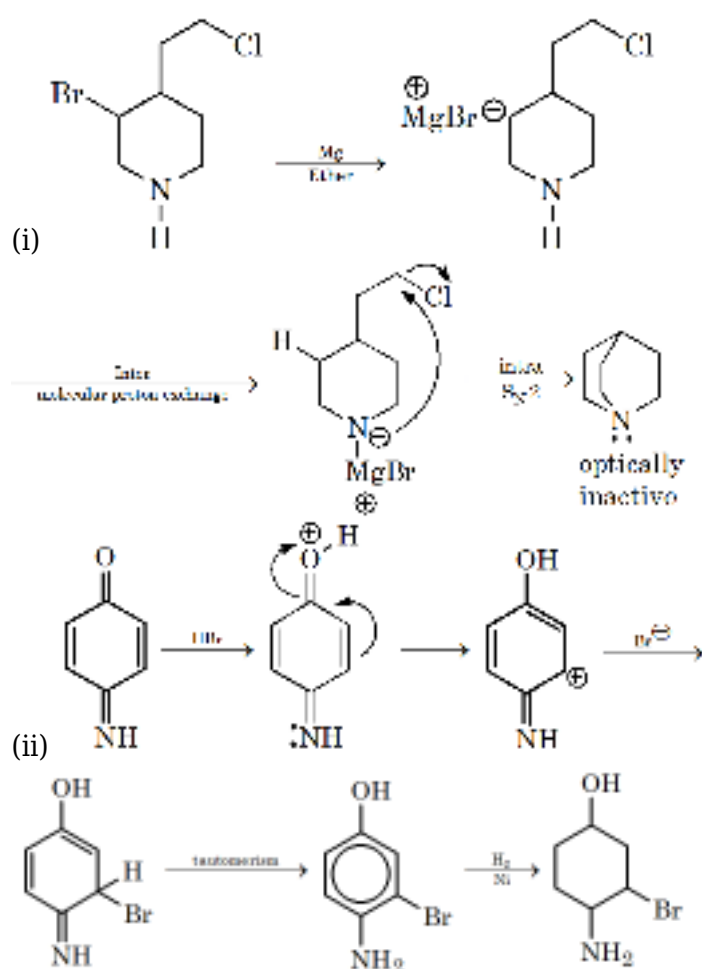
Aspartame → artificial sweetening agent

Final Answer: (B)

44) **Explain Question:** Identify which reaction(s) produce(s) single optically active Amine molecule

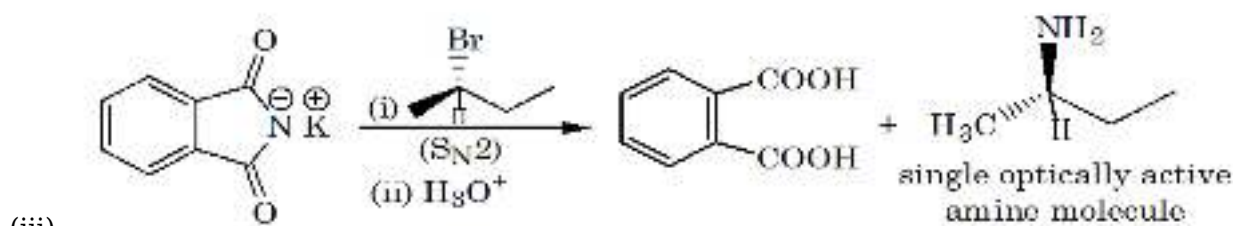
Concept: Preparation of aromatic & cyclic amines.

Solution:



generated.

3 chiral centres are



Final Answer: (B)

45) **Explain Question:** Identify compound which undergoes decarboxylation easily.

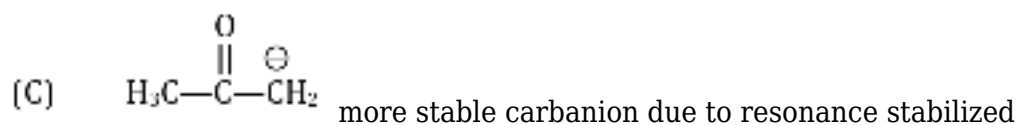
Concept: More stable carbanion $\rightarrow$ more rate of decarboxylation

Solution:

More stable carbanion $\rightarrow$ more rate of decarboxylation

$$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{COOH}$$

β - keto acid



Hence (C) undergo decarboxylation easily

Final Answer: (C)

46) **Explain Question:** Calculate the asking parameters for given complex.

Concept: Concept of co-ordination compounds.

Solution:

$$x = +2$$

$$y = +4$$

$$z = +3$$

$$x + y + z = 2 + 4 + 3 = 9$$

Final Answer: (9)

47) **Explain Question:** Calculate the % of metal exist as Fe^{+3}

Concept: Concept of stiochiometry & balancing.

Solution:



$$2 \left(\frac{100-x}{100} \times 0.98 \right) + 3 \left(\frac{x}{100} \times 0.98 \right) - 2 \times 1 = 0$$

$$200 - 2x + 3x$$

$$200 + x = \frac{200}{0.98}$$

$$x = 4.08\%$$

Final Answer: (4)

48) **Explain Question:** Calculate the final temperature for given process.

Concept: Apply concept of Irreversible adiabatic expansion.

Solution:

$$1 \times \frac{3}{2}R(T_2 - 300) = -P_{\text{ext}} \left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1} \right)$$

$$1.5(T_2 - 300) = -1 \left(\frac{T_2}{2} - \frac{300}{5} \right)$$

$$1.5T_2 - 450 = -0.5T_2 + 60$$

$$2T_2 = 510$$

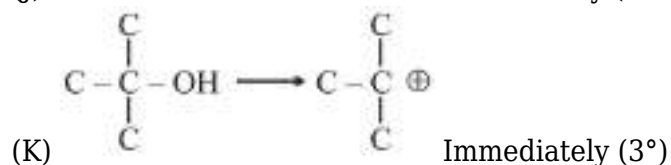
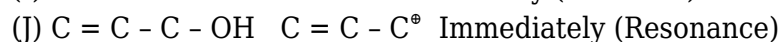
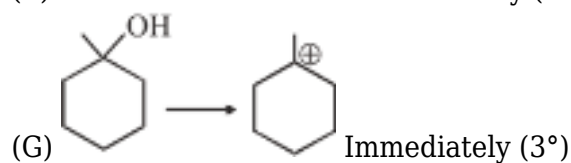
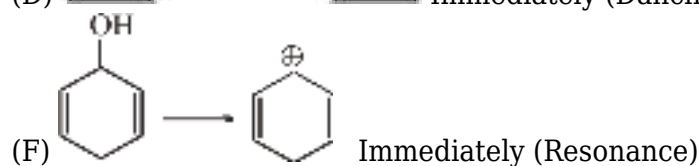
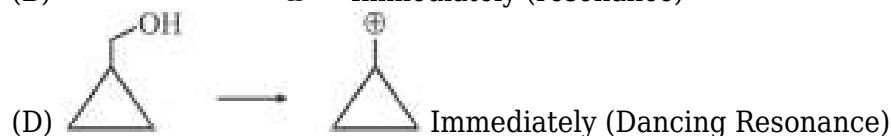
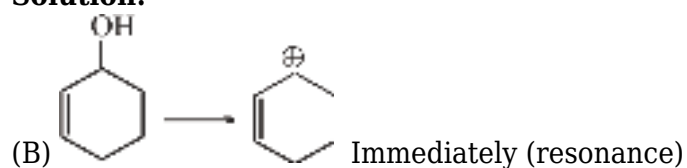
$$T_2 = 255 \text{ K}$$

Final Answer: (255)

49) **Explain Question:** Which compounds will give immediate turbidity with Lucas reagent

Concept: Turbidity with lucas reagent.

Solution:



Final Answer: (7)

50) **Explain Question:** Identify compounds gives positive test for nitrogen in lassaigen's test

Concept: Compounds having carbon, nitrogen, sulphur & halogens gives positive lassaigenes test.

Solution:

(iii), (iv) will give positive test.

Final Answer: (2)

51) **Explain Question :** We have to solve for x.

Concept : Types of trigonometric equations.

Solution :

Let $3^x \pi = \theta$ then

$$\begin{aligned}
 &\Rightarrow \cos \theta - 2 \cos^2 \theta + 2 \cos 4\theta - \cos 7\theta \\
 &= \sin \theta + 2 \sin^2 \theta - 2 \sin 4\theta + 2 \sin 3\theta - \sin 7\theta \\
 &\Rightarrow \cos \theta - \cos 7\theta + \sin 7\theta - \sin \theta \\
 &\quad + 2(\cos 4\theta + \sin 4\theta) = 2 + 2 \sin 3\theta \\
 &\Rightarrow 2 \sin 4\theta \sin 3\theta + 2 \cos 4\theta \sin 3\theta \\
 &\quad + 2(\cos 4\theta + \sin 4\theta) = 2 + 2 \sin 3\theta \\
 &\Rightarrow \sin 4\theta (\sin 3\theta + 1) + \cos 4\theta (\sin 3\theta + 1) \\
 &\quad - (1 + \sin 3\theta) = 0 \\
 &\Rightarrow (\sin 3\theta + 1)(\sin 4\theta + \cos 4\theta - 1) = 0 \\
 &\Rightarrow x = \log_3 \frac{4n-1}{2} - 1 \text{ And } \sin 4\theta + \cos 4\theta = 1 \\
 &\Rightarrow \sin \left(4\theta + \frac{\pi}{4} \right) = \sin \frac{\pi}{4} \Rightarrow 4\theta + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4} \\
 &\Rightarrow x = \log_3 \left(\frac{n}{4} + (-1)^n \frac{\pi}{16} - \frac{\pi}{16} \right), n \in \mathbb{Z}
 \end{aligned}$$

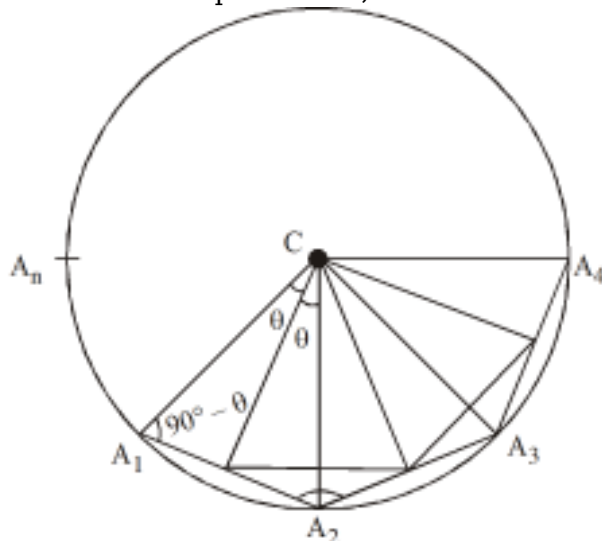
Final Answer : (A)

52) **Explain Question :** We have to find the value of the summation.

Concept : Sum of infinite G.P.

Solution :

The sum of the perimeters, then is



$$S = nx + nx \cos \theta + nx \cos^2 \theta + \dots = \frac{nx}{1 - \cos \theta} \quad (\text{where } \theta = \frac{\pi}{n})$$

$$\text{Put } n = 6, \quad \frac{6x}{1 - \cos \frac{\pi}{6}} = \frac{12x}{2 - \sqrt{3}} = 12$$

$$(\text{where } x \text{ is } \frac{1}{2 + \sqrt{3}} = 2 - \sqrt{3})$$

$$\cos(\pi - 2\theta) = \frac{\frac{x^2}{4} + \frac{x'^2}{4} - x'^2}{2 \left(\frac{x}{2}\right) \left(\frac{x}{2}\right)}$$

$$\frac{-x^2}{2} \cos 2\theta = \frac{x^2}{2} - x'^2$$

$$x'^2 = \frac{x^2}{2} (1 + \cos 2\theta) = x^2 \cos^2 \theta$$

$$\square \quad x' = x \cos \theta.$$

Final Answer : (D)

53) **Explain Question :** We have to solve for f(x).

Concept : Types of functions.

Solution :

$$f(x) = \int_0^1 \frac{dt}{f(xt)} \quad xt = z \quad dt = \frac{dz}{x}$$

$$f(x) = \frac{1}{x} \int_0^x \frac{dz}{f(z)} \Rightarrow xf(x) = \int_0^x \frac{dz}{f(z)}$$

$$xf'(x) + f(x) = \frac{1}{f(x)}$$

$$xf'(x) = \frac{1 - (f(x))^2}{f(x)}$$

$$\frac{2f(x)f'(x)}{1 - (f(x))^2} = \frac{2}{x}$$

$$\text{Integrate} - \log(1 - f(x)^2) = 2 \log x + c$$

$$f(1) = 0 \quad c = 0$$

$$\log(1 - f(x)^2) = \log \frac{1}{x^2}$$

$$f(x) = \sqrt{1 - \frac{1}{x^2}}$$

Final Answer : (D)

54) **Explain Question :** We have to find solution of the given DE.

Concept : Linear DE

Solution :

$$\frac{dy}{dx} = \frac{\sin x + y \cos x}{\sin x \cdot \cos x \left(\frac{1}{\cos x} - \sin x \cdot \frac{\sin x}{\cos x} \right)}$$

$$= \frac{\sin x + y \cos x}{\sin x (1 - \sin^2 x)}$$

$$\frac{dy}{dx} = \sec^2 x + y \cdot 2 (\operatorname{cosec} 2x)$$

$$\frac{dy}{dx} - 2 \operatorname{cosec} (2x) \cdot y = \sec^2 x$$

$$\frac{dy}{dx} + p \cdot y = Q$$

$$\text{I.F.} = e^{\int p dx} = e^{\int -2 \operatorname{cosec} (2x) dx}$$

$$\text{Let } 2x = t$$

$$2 \frac{dx}{dt} = 1$$

$$dx = \frac{dt}{2}$$

$$= e^{-\int \operatorname{cosec}(t) dt}$$

$$= e^{-\ln \left| \tan \frac{t}{2} \right|}$$

$$= e^{-\ln |\tan x|} = \frac{1}{|\tan x|}$$

$$y(\text{IF}) = \int Q(\text{IF}) dx + c$$

$$\Rightarrow y \frac{1}{|\tan x|} = \int \sec^2 x \cdot \frac{1}{|\tan x|} + c$$

$$y \cdot \frac{1}{|\tan x|} = \int \frac{dt}{|t|} + c$$

$$\text{for } \tan x = t$$

$$y \cdot \frac{1}{|\tan x|} = \ln |t| + c$$

$$y = |\tan x| (\ln |\tan x| + c)$$

$$\text{Put } x = \frac{\pi}{4}, y = 2$$

$$2 = \ln 1 + c \Rightarrow c = 2$$

$$y = |\tan x| (\ln |\tan x| + 2)$$

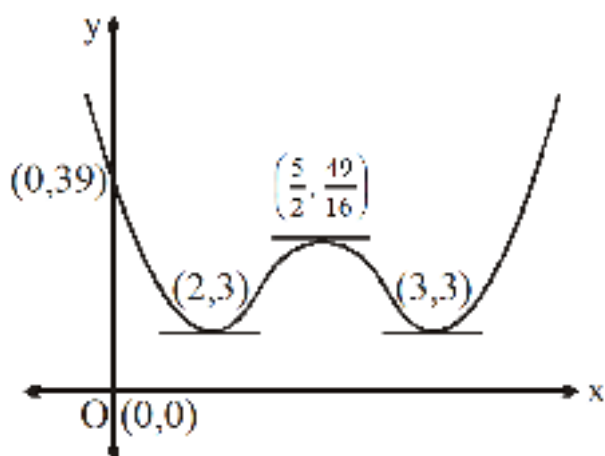
$$y\left(\frac{\pi}{3}\right) = \sqrt{3} (\ln \sqrt{3} + 2)$$

Final Answer : (A)

55) **Explain Question :** We have to solve the integrant and then find maxima.

Concept : Properties of DI and maxima - minima.

Solution :



$$h'(x) = f(x)$$

$$\int h(x) dx = g(x)$$

$$h(x) = g'(x) \quad h'(x) = g''(x) = f(x)$$

$$\int_{-\pi/2}^{\pi/2} (g(x) + g''(x)) \cos x \, dx = 5$$

□ Apply IBP

$$\int_{-\pi/2}^{\pi/2} g(x) \cos x \, dx + \int_{-\pi/2}^{\pi/2} g''(x) \cos x \, dx = 5$$

$$\int_{-\pi/2}^{\pi/2} g(x) \cos x \, dx + (\cos x \, g'(x))_{-\pi/2}^{\pi/2} + \int_{-\pi/2}^{\pi/2} \sin x \, g'(x) \, dx = 5$$

$$\Rightarrow \int_{-\pi/2}^{\pi/2} g(x) \cos x \, dx + (\sin x \, g(x))_{-\pi/2}^{\pi/2} - \int_{-\pi/2}^{\pi/2} \cos x \, g(x) \, dx = 5$$

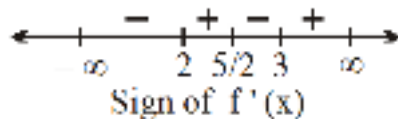
$$\Rightarrow g\left(\frac{\pi}{2}\right) + g\left(-\frac{\pi}{2}\right) = 5 \Rightarrow 3 + g\left(-\frac{\pi}{2}\right) = 5 \Rightarrow g\left(-\frac{\pi}{2}\right) = 2$$

Let $p(x) = k(x-2)^2(x-3)^2 + 3$

As, $p(1) = 4k + 3 = 7$

□ $k = 1 \Rightarrow f(x) = (x-2)^2(x-3)^2 + 3$

Now, $p'(x) = 2(x-2)(x-3)(2x-5)$



$\Rightarrow p(x)$ has local maximum at $x = \frac{5}{2}$

□ $p_{\max} \left(x = \frac{5}{2} \right) = \frac{49}{16}$

Final Answer : (B)

56) **Explain Question :** We have to solve the integrant and find the required area.

Concept : Properties of DI and area between curves.

Solution :

$$f(x) = 6(\ln x)^2 - 1 \int_1^e \frac{f(t)}{t} dt$$

$$\left(\int_1^e \frac{f(t)}{t} dt = \alpha \right)$$

$f(x) = 6(\ln x)^2 - \alpha \dots (1)$

$$\text{Now, } \alpha = 1 \int_1^e \frac{f(t)}{t} dt$$

$$= \int_1^e \frac{6(\ln t)^2 - \alpha}{t} dt$$

$$= 6 \int_1^e \frac{(\ln t)^2}{t} dt - \alpha \cdot \int_1^e \frac{1}{t} dt$$

$$= 6 \cdot \left(\frac{(\ln t)^3}{3} \right)_1^e - \alpha \cdot (\ln t)_1^e$$

$$= 2 \cdot (1^3 - 0^3) - \alpha(1 - 0)$$

$$= 2 - \alpha$$

$$\Rightarrow \alpha = 1$$

$$\text{So, } f(x) = 6 (\ln x)^2 - 1$$

$$\text{Let } y = g(x) = f(x) + 1 = 6 (\ln x)^2$$

$$g'(x) = 6 \cdot 2 \cdot (\ln x) \cdot \frac{1}{x} = 12 \cdot \left(\frac{\ln x}{x} \right)$$

$$g\left(\frac{1}{e}\right) = 12 \cdot \frac{\ln\left(\frac{1}{e}\right)}{\frac{1}{e}} = 12e \cdot (-1) = -12e$$

$$g'\left(\frac{1}{e}\right) = 6 \cdot \ln\left(\frac{1}{e}\right)^2 = 6 \cdot (-\ln e)^2 = 6$$

$$P\left(\frac{1}{e}, 6\right)$$

$$T: y - 6 = -12e \left(x - \frac{1}{e}\right)$$

$$T: y - 6 = -12ex + 12$$

$$y + 12ex = 18$$

$$x = 0 \Rightarrow y = 18$$

$$y = 0 \Rightarrow x = \frac{3}{2e}$$

$$\Delta = \frac{1}{2} \times 18 \times \frac{3}{2e} = \frac{27}{2e}$$

$$\frac{2e\Delta}{3} = 9$$

$$\left[\frac{2e\Delta}{3} \right] = 9$$

Final Answer : (D)

57) **Explain Question :** We have to find the required images of points.

Concept : Reflection of a point.

Solution :

Let A and B are images of (5, 3) about line $y = x$ and x-axis

□ A(3, 5) and B(5, -3)

Equation joining AB is $y + 3 = -4(x - 5)$

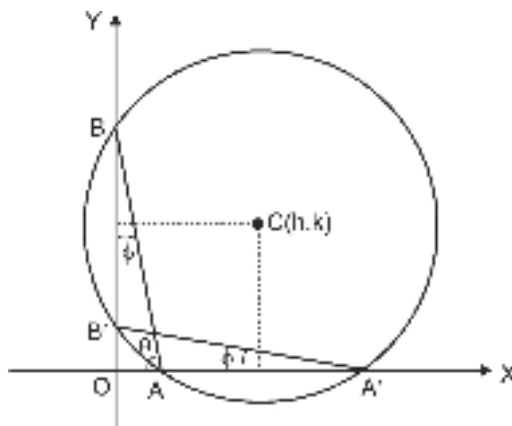
& is $\left(\frac{17}{4}, 0\right)$

Final Answer : (C)

58) **Explain Question :** We have to find the required locus.

Concept : Parametric coordinates.

Solution :



Let $\angle OA'B' = f$ and $\angle OAB = \phi$

$$\Rightarrow \theta + \phi = \frac{\pi}{2} \text{ and } \angle OBA = \phi$$

$\therefore$ length of AB is 'a' and length of A'B' is 'b'

□ from the figure

$$A'(b \cos \phi, 0) \text{ and } A(a \cos \theta, 0)$$

$$\text{similarly } B(0, a \sin \theta) \text{ and } B'(0, b \sin \phi)$$

Let $C(h, k)$ be the centre of circle

$$\square 2h = a \cos \theta + b \cos \phi$$

$$\therefore \theta = \frac{\pi}{2} - \phi$$

$$\square 2h = a \cos \theta + b \sin \theta \quad \dots\dots(i)$$

$$\text{and } 2k = a \sin \theta + b \sin \phi$$

$$\therefore \theta = \frac{\pi}{2} - \phi$$

$$\square 2k = a \sin \theta + b \cos \theta \quad \dots\dots(ii)$$

$$\text{on solving (i) and (ii), we get } \cos \theta = \frac{2ah - 2bk}{a^2 - b^2} \text{ and } \sin \theta = \frac{2ak - 2bh}{a^2 - b^2}$$

$$\therefore \sin^2 \theta + \cos^2 \theta = 1$$

$$\square \text{ locus of } C(h, k) \text{ is } (2ax - 2by)^2 + (2bx - 2ay)^2 = (a^2 - b^2)^2$$

Final Answer : (D)

59) **Explain Question :** We have to solve the given integrant.

Concept : Properties of DI and limits.

Solution :

$$\text{Given } f(x) = \lim_{n \rightarrow \infty} \frac{x^n - x^{-n}}{x^n + x^{-n}}; 0 < x < 1; n \in \mathbb{N}$$

$$\rightarrow 0 < x < 1, n \rightarrow \infty \Rightarrow x^n \rightarrow 0$$

$$1. f(x) = \lim_{n \rightarrow \infty} \frac{x^n - \frac{1}{x^n}}{x^n + \frac{1}{x^n}} = \frac{0 - \infty}{0 + \infty} = \lim_{n \rightarrow \infty} \frac{x^{2n} - 1}{x^{2n} + 1}$$

$$= -1$$

$$\rightarrow \int \sin^y(x) \cdot f(x) dx = \int -\sin^y x \cdot dx \times 1$$

$$= x \sin^y x - \int x \cdot \frac{1}{\sqrt{1-x^2}} dx$$

$$\text{Let } 1 - x^2 = t$$

$$x = \sqrt{1-t}$$

$$2x dx = dt$$

$$= x \sin^y x - \int \frac{(-1/2) dt}{\sqrt{t}}$$

$$= x \sin^{-1} x + \frac{1}{2} \frac{(+\sqrt{1-x^2})}{1/2} + C$$

$$= \left(x \sin^{-1} x + \sqrt{1-x^2} \right) + C$$

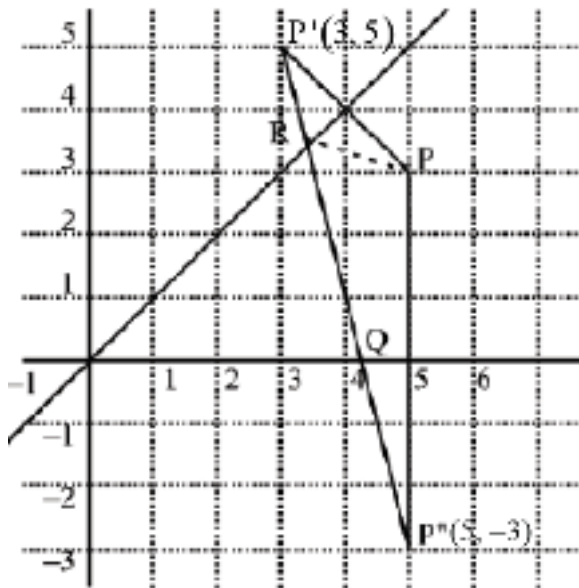
Multiply (-1) for $f(x) = - \left[x \sin^{-1} x + \sqrt{1-x^2} \right] + C$

Final Answer : (A)

60) **Explain Question :** We have to find Q.

Concept : Image of a point with line.

Solution :



P' and P'' are the images of P w.r.t. $y = x$ and $y = 0$ respectively. Therefore, $P' = (3, 5)$

The equation of $P'P''$ is

$$y + 3 = \frac{5 + 3}{3 - 5} (x - 5)$$

$$\therefore x = 5 + 3 \left(-\frac{1}{4} \right) = \frac{17}{4} \therefore Q \equiv \left(\frac{17}{4}, 0 \right)$$

Final Answer : (A)

61) **Explain Question :** We have to solve for $f_3(x)$

Concept : Summation of trigonometric series.

Solution :

$$f_1(x) = 2 \sum_{r=1}^n \frac{\sin \left(x + \frac{r\pi}{6} \right) - \left(x + (r-1) \frac{\pi}{6} \right)}{\cos \left(x + (r-1) \frac{\pi}{6} \right) \cdot \cos \left(x + \frac{r\pi}{6} \right)}$$

We have

$$= 2 \left[\left(\tan \left(x + \frac{\pi}{6} \right) - \tan x \right) + \left(\tan \left(x + \frac{2\pi}{6} \right) - \tan \left(x + \frac{\pi}{6} \right) \right) + \left(\tan \left(x + \frac{3\pi}{6} \right) - \tan \left(x + \frac{2\pi}{6} \right) \right) + \dots + \left(\tan \left(x + \frac{n\pi}{6} \right) - \tan \left(x + (n-1) \frac{\pi}{6} \right) \right) \right]$$

$$\Rightarrow f_1(x) = 2 \left(\tan \left(x + \frac{n\pi}{6} \right) - \tan x \right)$$

$$\text{For } n = 3, f_1(x) = 2 \left(\tan \left(\frac{\pi}{2} + x \right) - \tan x \right) = 2 (-\cot x - \tan x) = -2 \frac{1}{\sin x \cos x}$$

$$\Rightarrow 2f_2(x) = f_1(x) - 2 \tan \left(x + \frac{n\pi}{6} \right)$$

$$= 2 \tan \left(x + \frac{n\pi}{6} \right) - 2 \tan x - 2 \tan \left(x + \frac{n\pi}{6} \right)$$

$$\therefore f_2(x) = -\tan x$$

$$\Rightarrow f_3(x) = -f_2(x), \text{ so } \Rightarrow f_3(x) = \tan x$$

Final Answer : (C)

62) **Explain Question :** We have to solve the limit and examine the discontinuity.

Concept : Continuity at a point.

Solution :

$$(A) \quad f(1^+) = \lim_{h \rightarrow 0} \left(\lim_{n \rightarrow \infty} \frac{\log(2 + 1 + h) - (1 + h)^{2n} \sin(1 + h)}{1 + (1 + h)^{2n}} \right)$$

$$\lim_{h \rightarrow 0} \left(\lim_{n \rightarrow \infty} \frac{\frac{\ln(3+h)}{(1+h)^{2n}} - \sin(1 + h)}{\frac{1}{(1+h)^{2n}} + 1} \right)$$

$$= -\sin 1$$

$$f(1^-) = \lim_{h \rightarrow 0} \left(\lim_{n \rightarrow \infty} \frac{\ln(2 + 1 - h) - (1 - h)^{2n} \sin(1 - h)}{1 + (1 - h)^{2n}} \right)$$

$$= \frac{\ln 3 - 0}{1 + 0} = \ln 3$$

$$f(1) = \frac{\ln 3 - \sin 1}{2}$$

discontinuous at $x = 1$

$$(B) \quad f(x) = \lim_{n \rightarrow \infty} \frac{\log(2 + x) - x^{2n} \sin x}{1 + x^{2n}}$$

$$f(x) = \lim_{n \rightarrow \infty} \frac{\ln(2 - 1) - (-1)^{2n} \sin(-1)}{1 + (-1)^{2n}}$$

$$= \frac{0 + \sin 1}{2}$$

$$f(-1 + h) = \lim_{h \rightarrow 0} \left(\lim_{n \rightarrow \infty} \frac{\ln(2 - 1 + h) - (-1 + h)^{2n} (\sin(-1 + h))}{1 + (-1 + h)^{2n}} \right)$$

$$= \lim_{h \rightarrow 0} \left(\lim_{n \rightarrow \infty} \frac{\ln(1 + h) - (-1 + h)^{2n} (\sin(1 - h))}{1 + (-1 + h)^{2n}} \right)$$

$$= \lim_{h \rightarrow 0} \left(\frac{\ln(1 + h) + 0}{1 + 0} \right) = 0$$

$$= \lim_{h \rightarrow 0} \left(\lim_{n \rightarrow \infty} \frac{\ln(2 - 1 - h) - (-1 - h)^{2n} \sin(-1 - h)}{1 + (-1 - h)^{2n}} \right) = 0$$

$$= \lim_{h \rightarrow 0} \left(\frac{\frac{\ln(1-h)}{(1+h)^{2n}} + \sin(1 + h)}{\frac{1}{(1+h)^{2n}} + 1} \right)$$

$$= \sin 1$$

discontinuous at $x = -1, 1$

Final Answer : (C)

63) **Explain Question :** We have to solve the given limit.

Concept : Methods of finding limits.

Solution :

$$\left(1 + \frac{1}{n}\right)^{n+x_n} = e$$

Take ln on both sides

$$(n + x_n) \ln \left(1 + \frac{1}{n} \right) = 1$$

$$n + x_n = \frac{1}{\ln \left(1 + \frac{1}{n} \right)}$$

$$x_n = \frac{1}{\ln \left(1 + \frac{1}{n} \right)} - n$$

$$n = \frac{1}{t}$$

$$\lim_{n \rightarrow \infty} x_n = \lim_{t \rightarrow 0} \frac{1}{\ln(1+t)} - \frac{1}{t}$$

$$= \lim_{t \rightarrow 0} \frac{t - \ln(1+t)}{t \ln(1+t)}$$

using expansion

$$\lim_{t \rightarrow 0} \frac{t - \left[t - \frac{t^2}{2} + \frac{t^3}{3} \dots \right]}{t \left(t - \frac{t^2}{2} + \frac{t^3}{3} \dots \right)}$$

$$= \frac{1}{2}$$

Final Answer : (B)

64) **Explain Question :** We have to find the values of A and B.

Concept : Integration by substitution.

Solution :

$$\int \frac{\sin^3 x + \cos^3 x}{\sqrt{\sin^3 x \cos^3 x} \sin(x - \theta)} dx$$

$$I = \int \frac{\sin^3 x + \cos^3 x}{\sqrt{\sin^3 x \cos^3 x} (\sin x \cos \theta - \cos x \sin \theta)} dx$$

$$= \int \frac{\sin^3 x}{\sin^3 x \cos^2 x \sqrt{\tan x \cos \theta - \sin \theta}} dx + \int \frac{\cos^3 x}{\sin^2 x \cos^3 x \sqrt{\cos \theta - \cot x \sin \theta}} dx$$

$$= \int \frac{\sec^2 x}{\sqrt{\tan x \cos \theta - \sin \theta}} dx + \int \frac{\operatorname{cosec}^2 x}{\sqrt{\cos \theta - \cot x \sin \theta}} dx$$

$$I = I_1 + I_2 \quad \dots \text{[Let]}$$

$$\text{For } I_1, \text{ let } \tan x \cos \theta - \sin \theta = t^2$$

$$\sec^2 x dx = \frac{2t dt}{\cos \theta}$$

$$\text{For } I_2, \text{ let } \cos \theta - \cot x \sin \theta = z^2$$

$$\operatorname{cosec}^2 x dx = \frac{2z dz}{\sin \theta}$$

$$I = I_1 + I_2$$

$$= \int \frac{2t dt}{\cos \theta t} + \int \frac{2z dz}{\sin \theta z}$$

$$= \frac{2t}{\cos \theta} + \frac{2z}{\sin \theta}$$

$$= 2 \sec \theta \sqrt{\tan x \cos \theta - \sin \theta} + 2 \operatorname{cosec} \theta \sqrt{\cos \theta - \cot x \sin \theta}$$

Comparing

$$AB = 8 \operatorname{cosec} 2\theta$$

Final Answer : (D)

65) **Explain Question :** We have to solve the integrand and find $f(x)$.

Concept : Properties of DI

Solution :

$$e^x \left(\int_0^x f(t) dt + \int_0^x t f(x-t) dt + 1 \right) = 1 \int_0^x f(t) dt + \int_0^x t f(x-t) dt + 1 = e^{-x}$$

$$\int_0^x f(t) dt + \int_0^x (x-t) f(t) dt + 1 = e^{-x}$$

$$\int_0^x f(t) dt + x \int_0^x f(t) dt - \int_0^x t f(t) dt + 1 = e^{-x} \quad \dots(1)$$

$$f(x) + xf(x) + \int_0^x f(t) dt \cdot 1 - xf(x) = -e^{-x}$$

$$(x+1)f(x) + \int_0^x f(t) dt = xf(x) - e^{-x} \quad \dots(1)$$

$$(x+1)f'(x) + f(x) + \cancel{f(x)} = xf'(x) + \cancel{f(x)} + e^{-x}$$

$$f'(x) + f(x) = e^{-x}$$

$$(e^x f(x))' = 1 \Rightarrow e^x f(x) = x + c$$

$$\text{in (1) put } x = 0 \quad f(0) = -1$$

$$\therefore f(0) = c \Rightarrow c = -1$$

$$e^x f(x) = x - 1 \Rightarrow f(x) = \frac{x-1}{e^x}$$

$$f(2) = \frac{1}{e^2}$$

Final Answer : (B)

66) **Explain Question :** We have to find the required locus.

Concept : Radical axis.

Solution :

$$c_1 - c_2 = 0$$

$$\text{gives } x = 0.$$

$$\text{So } A(0, \sqrt{13}) \text{ and } B(0, -\sqrt{13})$$

Let

$$AP : (y - \sqrt{13}) \cos \theta = x \sin \theta$$

$$BQ : (y + \sqrt{13}) \cos \theta = x \sin \theta$$

Hence

$$P(-\sqrt{13} \sin 2\theta - 2 \cos 2\theta - 2, \sqrt{13} \cos 2\theta - 2 \sin 2\theta)$$

$$Q(\sqrt{13} \sin 2\theta + 4 \cos 2\theta + 4, -\sqrt{13} \cos 2\theta + 4 \sin 2\theta)$$

So

$$M = \frac{P+Q}{2} = (\cos 2\theta + 1, \sin 2\theta)$$

locus is circle: $(x-1)^2 + y^2 = 1$

Final Answer : (D)

67) **Explain Question :** We have to use integration and find relation between $f(x)$ and $g(x)$.

Concept : Properties of integration.

Solution :

$$f''(x) = g''(x) + 6x \quad \dots(1)$$

$$f'(1) = 4g'(1) - 3 = 9 \quad \dots(2)$$

$$f(2) = 3g(2) = 12 \quad \dots(3)$$

By integrating (1)

$$f'(x) = g'(x) + 6\frac{x^2}{2} + C$$

At $x = 1$,

$$f'(1) = g'(1) + 3 + C$$

$$\Rightarrow 9 = 4 + 3 + C \Rightarrow C = 3$$

$$\square f'(x) = g'(x) + 3x^2 + 3$$

Again by integrating,

$$f(x) = g(x) + \frac{3x^3}{3} + 3x + D$$

At $x = 2$,

$$f(2) = g(2) + 8 + 3(2) + D$$

$$\Rightarrow 12 = 4 + 8 + 6 + D \Rightarrow D = -6$$

$$\text{So, } f(x) = g(x) + x^3 + 3x - 6$$

$$\Rightarrow f(x) - g(x) = x^3 + 3x - 6$$

At $x = -2$,

$$\Rightarrow g(-2) - f(-2) = 20 \text{ (Option (A) is true)}$$

Now, for $-1 < x < 2$

$$h(x) = f(x) - g(x) = x^3 + 3x - 6$$

$$\Rightarrow h'(x) = 3x^2 + 3$$

$$\Rightarrow h(x) \uparrow$$

$$\text{So, } h(-1) < h(x) < h(2)$$

$$\Rightarrow -10 < h(x) < 8$$

$$\Rightarrow |h(x)| < 10 \text{ (option (B) is NOT true)}$$

$$\text{Now, } h'(x) = f'(x) - g'(x) = 3x^2 + 3$$

$$\text{If } |h'(x)| < 6 \Rightarrow |3x^2 + 3| < 6$$

$$\Rightarrow 3x^2 + 3 < 6$$

$$\Rightarrow x^2 < 1$$

$$\Rightarrow -1 < x < 1 \text{ (option (C) is True)}$$

$$\text{If } x \in (-1, 1) |f'(x) - g'(x)| < 6$$

option (C) is true and now to solve

$$f(x) - g(x) = 0$$

$$\Rightarrow x^3 + 3x - 6 = 0$$

$$h(x) = x^3 + 3x - 6$$

$$\text{here, } h(1) = -ve \text{ and } h\left(\frac{3}{2}\right) = +ve$$

So there exists $x_0 \in \left(1, \frac{3}{2}\right)$ such that $f(x_0) = g(x_0)$

Final Answer : (B)

68) **Explain Question :** We have to find 'a' by finding f(x).

Concept : Properties of ITF.

Solution :

$$\text{Clearly, } f(x) = \frac{\tan^{-1} x}{1 + \frac{2}{\pi} \tan^{-1} x}$$

Now domain of equation $f^2(x) + (\sin^{-1} x)^2 = a$, is $x \in [-1, 1]$

$$\text{Now, } f(x) = \frac{\pi}{2} \left(\frac{\frac{2}{\pi} \tan^{-1} x + 1 - 1}{1 + \frac{2}{\pi} \tan^{-1} x} \right)$$

$$= \frac{\pi}{2} \left(1 - \frac{1}{1 + \frac{2}{\pi} \tan^{-1} x} \right)$$

$$\text{So, } f(x)|_{\min} = \frac{\pi}{2} \text{ at } x = -1$$

$$f(x)|_{\max} = \frac{\pi}{6} \text{ at } x = 1$$

$$\text{So, } (f(x))^2 \in \left[0, \frac{\pi^2}{4}\right] \text{ and minimum and maximum is attained at } x = 0 \text{ and } x = -1$$

$$\text{Also, } (\sin^{-1} x)^2 \in \left[0, \frac{\pi^2}{4}\right]$$

$$\therefore a_{\max} = \frac{\pi^2}{2} \text{ at } x = -1 \text{ and } a_{\min} = 0 \text{ at } x = 0$$

$$\text{So, } a \in \left[0, \frac{\pi^2}{2}\right]$$

$$\therefore a_{\text{integer}} = 0, 1, 2, 3, 4$$

Hence, sum of integral values of a = 10.

Final Answer : (D)

69) **Explain Question :** We have to solve for x and y.

Concept : Product and sum formula.

Solution :

$$x = \cos 1^\circ \cos 89^\circ \cos 2^\circ \cos 88^\circ \cos 3^\circ \cos 87^\circ \dots$$

$$x = \cos 1^\circ \cos(90^\circ - 1^\circ) \cos 2^\circ \cos(90^\circ - 2^\circ) \cos 3^\circ \cos(90^\circ - 3^\circ) \dots$$

$$x = \cos 1^\circ \sin 1^\circ \cos 2^\circ \sin 2^\circ \dots \cos 44^\circ \sin 44^\circ \cos 45^\circ$$

Divide and multiply by 2^{44}

$$x = \frac{1}{\sqrt{2}} \frac{2^{44}}{2^{44}} \cos 1^\circ \sin 1^\circ \cos 2^\circ \sin 2^\circ \dots$$

$$x = \frac{1}{\sqrt{2} \times (2^{44})} (\sin 2^\circ \sin 4^\circ \dots \sin 88^\circ)$$

$$x = \frac{1}{\sqrt{2} 2^{44}} \sin 2^\circ \cos 2^\circ \sin 4^\circ \cos 4^\circ \dots \sin 44^\circ \cos 44^\circ$$

Divide and multiply by 2^{22}

$$\begin{aligned}
 x &= \frac{1}{\sqrt{2}(2^{44} \times 2^{22})} \sin 4^\circ \sin 8^\circ \sin 12^\circ \dots \sin 88^\circ \\
 x &= \frac{1}{2^{66.5}} \cos 2^\circ \cos 6^\circ \dots \cos 86^\circ \\
 x &= \frac{y}{2^{66.5}} \\
 \frac{2}{7} \log_2 \left(\frac{y}{x} \right) &= \frac{2}{7} \times \log_2 2^{66.5} \\
 &= \frac{2}{7} \times 66.5 \\
 &= 19
 \end{aligned}$$

Final Answer : (C)

70) **Explain Question :** We have to solve the summation.

Concept : Special sequences.

Solution :

$$\begin{aligned}
 a_{i+1} &= \frac{-a_i + a_i \sqrt{4a_i - 3}}{2} \\
 \Rightarrow (a_i)^3 &= (a_{i+1})^2 + (a_i)^2 + a_i a_{i+1}
 \end{aligned}$$

Hence

$$\begin{aligned}
 \sum_{i=1}^{2023} \frac{(a_{i+1}^3 - a_i^3 + a_i^3)}{a_i^2 + a_{i+1}^2 + a_i a_{i+1}} &= \sum_{i=1}^{2023} (a_{i+1} - a_i) + 2023 \\
 &= a_{2024} - a_1 + 2023 = 4047 - a_1
 \end{aligned}$$

Final Answer : (B)

71) **Explain Question :** We have to solve the given expression and find λ .

Concept : Inverse of a function.

Solution :

$$\begin{aligned}
 (x+1)^3 + (y+1)^3 &= 16 \\
 y &= -1 + \left(16 - (x+1)^3\right)^{\frac{1}{3}} = f(x) \\
 x &= -1 + \left(16 - (y+1)^3\right)^{\frac{1}{3}} \\
 \therefore f^{-1}(x) &= -1 + \left(16 - (x+1)^3\right)^{1/3} = g(x) \\
 \Rightarrow f(x) &= g(x) \\
 f\left(-1 + \sqrt[3]{15}\right) &= g\left(-1 + \sqrt[3]{15}\right) = 0 \\
 (f(x) + g(x)).g(x + f(x)) &= (f(x + f(x)))^2 \\
 \frac{d}{dx}(f(x + f(x)))^2 &= 2f(x + f(x)) \\
 f'(x + f(x)).(1 + f'(x)) &= 0 \\
 \frac{d}{dx}(f(x + f(x)))^2 \text{ at } x = -1 + \sqrt[3]{15} &= 0 \\
 \therefore g\left(-1 + \sqrt[3]{15}\right) &= f\left(-1 + \sqrt[3]{15}\right) = 0
 \end{aligned}$$

Final Answer : (0)

72) **Explain Question :** We have to find the value of 'n'.

Concept : Problems based on nC_r .

Solution :

$a_{\frac{n}{2}}$ and $a_{\frac{(n+1)}{2}}$ are both 1, Note that the number of ways to choose numbers left of $a_{\frac{n}{2}}$ and

$a_{\frac{n+1}{2}}$ $\left(a_1 > a_2 > a_3 \dots a_{\frac{n}{2}}; a_{\frac{n+1}{2}} \right)$ is same as that of right. We started with 9

$$n = 9 \Rightarrow {}^8C_4 = 70 \text{ (rejected)}$$

$$n = 10 \Rightarrow {}^9C_4 = 126 \text{ (rejected)}$$

$$n = 11 \Rightarrow {}^{10}C_3 = 252$$

$$n = 12 \Rightarrow {}^{11}C_5 = 462$$

$$n = 13 \Rightarrow {}^{12}C_6 = 924 \text{ (rejected)}$$

Sum of possible values of n is $(11 + 12) = 23$

Final Answer : (3)

73) **Explain Question :** We have to find n.

Concept : Special sequences.

Solution :

We have $T_r - 3T_{r-1} = 6^r$

Dividing both sides by 3^r , we get

$$\frac{T_r}{3^r} - \frac{T_{r-1}}{3^{r-1}} = 2^r \Rightarrow \sum_{r=2}^n \left(\frac{T_r}{3^r} - \frac{T_{r-1}}{3^{r-1}} \right) = \underbrace{\sum_{r=2}^n 2^r}_{\text{G.P.}}$$

$$\left. \begin{array}{l} \frac{T_2}{3^2} - \frac{T_1}{3} \\ + \frac{T_3}{3^3} - \frac{T_2}{3^2} \\ + \frac{T_4}{3^4} - \frac{T_3}{3^3} \\ \vdots \\ + \frac{T_n}{3^n} - \frac{T_{n-1}}{3^{n-1}} \end{array} \right\} = \underbrace{\sum_{r=2}^n 2^r}_{\text{G.P.}}$$

$$\frac{T_n}{3^n} - \frac{T_1}{3} = 4(2^{n-1} - 1)$$

$$\frac{T_n}{3^n} - \frac{6}{3} = 4(2^{n-1} - 1) \quad (\because T_1 = 6)$$

$$\Rightarrow T_n = 2 \cdot 3^n + 4 \cdot 3^n \cdot 2^{n-1} - 4 \cdot 3^n = 2^{n+1} \cdot 3^n - 2 \cdot 3^n$$

$$\Rightarrow T_n = 2(6^n - 3^n)$$

$$S_n = 2 \left(\sum_1^n 6^n - \sum_1^n 3^n \right) = 2 \left[\frac{6}{5} (6^n - 1) - \frac{3}{2} (3^n - 1) \right]$$

So,

$$\Rightarrow S_n = \frac{3}{5} [4 \cdot 6^n - 5 \cdot 3^n + 1]$$

Now on comparing, we get

$$\frac{(n^2 - 12n + 39)(4 \cdot 6^n - 5 \cdot 3^n + 1)}{5} = \frac{3}{5} [4 \cdot 6^n - 5 \cdot 3^n + 1]$$

$$\Rightarrow n^2 - 12n + 39 = 3 \Rightarrow (n - 6)^2 = 0$$

Hence $n = 6$ Ans.]

Final Answer : (6)

74) **Explain Question :** We have to find the values of 'm' and 'n'.

Concept : Properties of integration.

Solution :

$$I = \int \underbrace{\frac{\cot 2x \cot 3x}{\cot 5x}}_{I_1} dx - \int \underbrace{\tan 2x \tan 5x \tan 7x}_{I_2} dx$$

Consider $I_2 = \int (\tan 7x - \tan 5x - \tan 2x) dx$

Again $I_1 = \int \frac{\cot 2x \cot 3x}{\cot 5x} dx$

$5x = 2x + 3x;$

$\cot 5x = \cot(A+B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$

$\cot 5x = \frac{\cot 2x \cot 3x - 1}{\cot 2x + \cot 3x}$

$\cot 5x \cot 2x + \cot 5x \cot 3x + 1 = \cot 2x \cot 3x$

$I_1 = \int \frac{\cot 5x \cot 2x + \cot 5x \cot 3x + 1}{\cot 5x} dx$

$I_1 = \int (\cot 2x + \cot 3x + \tan 5x) dx$

Now, $I = I_1 - I_2$

$I = \int (\cot 2x + \cot 3x + \tan 5x) dx - \int (\tan 7x - \tan 5x - \tan 2x) dx$

$= \int (\cot 2x + \tan 2x) dx + \int (\cot 3x) dx + 2 \int (\tan 5x) dx - \int (\tan 7x) dx$

$= 2 \int (\operatorname{cosec} 4x) dx + \frac{\ln(\sin 3x)}{3} + \frac{2}{5} \ln(\sec 5x) + \frac{1}{7} \ln(\cos 7x)$

$= \frac{1}{2} \ln(\operatorname{cosec} 4x - \cot 4x) + \frac{\ln(\sin 3x)}{3} + \frac{2}{5} \ln(\sec 5x) + \frac{1}{7} \ln(\cos 7x) + C$

$= \frac{1}{2} \ln(\tan 2x) + \frac{1}{3} \ln(\sin 3x) + \frac{2}{5} \ln(\sec 5x) + \frac{1}{7} \ln(\cos 7x) + C$

Hence, $a + b + c + d = \frac{1}{2} + \frac{1}{3} + \frac{2}{5} + \frac{1}{7}$

$= \frac{105 + 70 + 84 + 30}{210} = \frac{289}{210} = \frac{m}{n} \Rightarrow m + n = 499$

Final Answer : (499)

75) **Explain Question :** We have to solve integration and find f(x).

Concept : Properties of integration.

Solution :

$9x^4 - 4\sqrt{3}x + 3 = (3x^2)^2 - 2 \times 3x^2 + 1 + 6x^2 - 4\sqrt{3}x + 2$

$= (3x^2 - 1)^2 + 2(3x^2 - 2\sqrt{3}x + 1)$

$= (\sqrt{3}x - 1)^2 \left[(\sqrt{3}x + 1)^2 + 2 \right]$

$\frac{1}{6} \int \frac{(4 + 6x^2) dx}{\left((\sqrt{3}x + 1)^2 + 2 \right) (\sqrt{3}x - 1)^2} = \frac{1}{6} \int \frac{(\sqrt{3}x + 1)^2 + 2 + (\sqrt{3}x - 1)^2}{\left[(\sqrt{3}x + 1)^2 + 2 \right] [\sqrt{3}x - 1]^2} dx$

$$\begin{aligned}
&= \frac{1}{6} \int \frac{dx}{(\sqrt{3}x+1)^2 + 2} + \frac{1}{6} \int \frac{dx}{(\sqrt{3}x-1)^2} \\
&= \frac{1}{6} \cdot \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{3}x+1}{\sqrt{2}} \right) \cdot \frac{1}{\sqrt{3}} + \frac{1}{6} \left(\frac{-1}{\sqrt{3}x-1} \right) \left(\frac{+1}{\sqrt{3}} \right) + C \\
&= \frac{1}{6\sqrt{6}} \tan^{-1} \left(\frac{\sqrt{3}x+1}{\sqrt{2}} \right) - \frac{1}{6\sqrt{3}} \cdot \frac{1}{(\sqrt{3}x-1)} + C
\end{aligned}$$

$$f(x) = \sqrt{3}x - 1$$

Final Answer : (2)